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by

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ABSTRACT

FilmTrove is an advanced movie database and recommendation management system tailored to deliver personalized film discoveries and organized access to movie-related metadata. The platform integrates core entities such as users, movies, cast, genres, ratings, languages, streaming availability, and watch history to foster an engaging and personalized viewing experience.

The system features robust modules for user registration, authentication, movie metadata entry, watch history logging, and review submission. It utilizes SQL-based queries in place of a traditional recommendation engine to provide real-time, tailored recommendations based on user preferences and viewing behavior. Admin users maintain control over database integrity through movie, user, and review moderation panels.

By offering multilingual and multi-platform support, FilmTrove makes it easier for users to find content across their preferred languages and streaming services. Backed by a relational schema normalized to BCNF and developed using Visual Studio with C# and SQLPlus, FilmTrove ensures optimal data integrity, performance, and scalability.

Keywords: Movie database, Watch history, SQL-based recommendations, Streaming platforms, Multilingual metadata, Genre-based filtering, User review system, Admin moderation, BCNF normalization, C# backend, SQLPlus integration, Movie analytics

SDG Goals Covered:

SDG 4: Quality Education

- Promotes media literacy by providing structured access to diverse global cinema and metadata.

SDG 9: Industry, Innovation, and Infrastructure

- Implements a scalable, digital movie infrastructure using normalized relational databases and modular architecture.

SDG 10: Reduced Inequalities

- Supports multilingual data and cross-cultural film discovery, ensuring inclusive access to global cinema content.

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Chapter 1

Introduction to FilmTrove – The Only Cinematic Compass You Will Ever Need

1.1 Introduction

FilmTrove is an intelligent, personalized movie recommendation and database system that revolutionizes the way users find and interact with movies. In the current rapid digital age, audiences frequently find it difficult to locate content that suits their individual tastes. FilmTrove addresses this issue by integrating a strong search engine, smart recommendation logic, and a comprehensive movie database to offer a one-stop solution for film enthusiasts.

Whether you're a casual viewer looking for your next watch or an admin managing user feedback and content, FilmTrove offers tailored experiences through an intuitive interface and advanced filtering capabilities. By capturing user ratings, watch history, genres, languages, and streaming platform availability, FilmTrove delivers personalized recommendations that evolve with user behavior.

Driven by the disconnect between the user expectations and the fractured landscape of current film discovery platforms, FilmTrove provides a single, smart system to make discovering movies easy, enjoyable, and data driven.

1.2 Three Tier Architecture

The system adopts a three-tier database architecture to promote modularity, scalability, and streamlined maintenance. At the external level, users — including movie enthusiasts, guests, and administrators — engage with FilmTrove through a dynamic and responsive web interface. This interface is designed for simplicity and personalization, allowing users to explore content, manage profiles, and access tailored recommendations. The conceptual level defines the logical structure and relationships between core entities such as User, Movie, Review, Watch History, Genre, Language, and Streaming Service. This layer forms the backbone of FilmTrove's functionality, ensuring consistent and meaningful data interactions. At the internal level, the system ensures efficient data storage, optimized query performance, and strict adherence to security and data integrity standards.

The core aim of this project is to redefine digital movie discovery by offering a centralized, intelligent, and scalable platform. By automating and optimizing operations like search, recommendation generation, content moderation, and user interaction tracking, FilmTrove

enhances user satisfaction, increases engagement, and provides admins with actionable insights — all while maintaining a high-performance, data-driven environment.

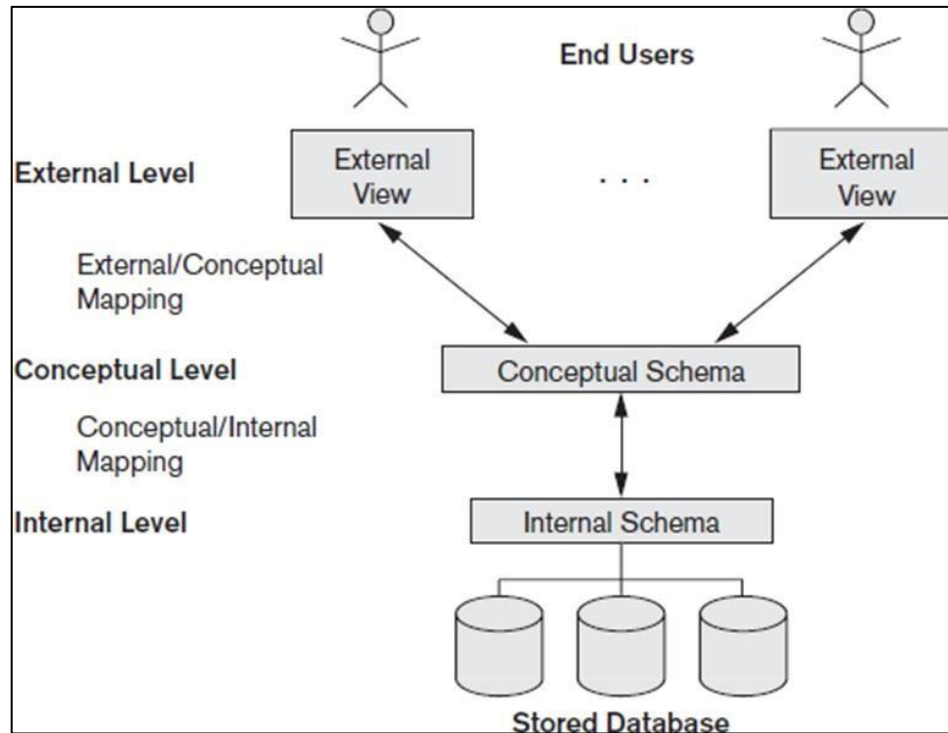


Figure 1.1: Three Tier Architecture

Chapter 2

Background

2.1 Introduction

In today's digital age flooded with content, the entertainment landscape has undergone a massive shift towards personalization and intelligent media management. With millions of viewers streaming, reviewing, and discovering movies on-line, the demand for scalable and high-performance movie database systems is stronger than ever before. Such systems are not only supposed to support a broad array of functionalities — ranging from user authentication and watch history to intelligent suggestions, streaming availability, multi-language support, and content moderation — but also do so under low-latency conditions at scale.

Though several movie-oriented apps and services have arisen to fill these needs, most remain inadequate — bound by fractured data, weak personalization, or inability to bend and adjust to users' likes.

2.2 Existing Systems

1. **Mainstream Streaming Platforms** – Services like Netflix, Prime Video, and Disney+ provide vast movie libraries with curated content, but they prioritize content delivery over database openness. Their recommendation systems are proprietary, offering little insight into how user preferences and metadata are stored, analysed, or optimized. This makes them unsuitable for academic exploration or structural customization.
2. **Movie Databases (e.g., IMDb, TMDb)** – Platforms like IMDb and TMDb offer rich metadata on movies, cast, ratings, and reviews. While TMDb does provide API access for development purposes, these systems are informational in nature and do not offer personalized recommendations or user watch history tracking. They lack integration with dynamic user interaction features like review management, secure login, or streaming redirects.
3. **Open-Source Media Management Tools** – Solutions like Jellyfin or Kodi support self-hosted media library management but are geared toward organizing existing offline collections rather than interacting with external APIs or offering cloud-based recommendation systems. These tools require significant setup and are not tailored for modular development or educational projects that emphasize relational database normalization, entity design, and UI integration.

2.3 Gaps Identified in Existing Systems

- **Disjointed Data Architecture:** Most current movie platforms keep data such as user profiles,

reviews, watch history, and streaming availability in separate silos, which hinders the implementation of unified, real-time personalization or analytics.

- **Lack of Personalization and Role-Based Access:** Most platforms provide generalized content feeds with minimal customization. Few offer dynamic recommendations based on user preferences or differentiated interfaces for admins, reviewers, or regular users.
- **Limited Educational Utility:** Most of the movie discovery or streaming tools are either commercial or closed source. Even open-source tools often require significant technical expertise and are not modular or intuitive enough for academic use or experimentation.

Chapter 3

Objectives

- ❖ **To design a relational and normalized database system** that captures detailed metadata about users, movies, genres, cast, streaming services, reviews, and watch history.
- ❖ **To implement a personalized recommendation engine** based on collaborative filtering using SQL queries, enhancing user engagement and satisfaction.
- ❖ **To offer powerful search and filtering tools** enabling users to browse movies by title, language, genre, rating, release year, or availability on streaming platforms.
- ❖ **To build role-based user interfaces** for regular users, guest users, and admins, each offering custom functionality and secure access.
- ❖ **To ensure high performance and data integrity** using third normal form (3NF) design, indexing strategies, and secure authentication protocols.
- ❖ **To maintain compliance with GDPR and other privacy regulations**, encrypting sensitive data and providing user control over data visibility and deletion.
- ❖ **To enable future scalability and integrations**, including support for new languages, third-party streaming APIs, and advanced features like live watch parties and notifications.

Software Requirements Specification

for

FilmTrove

Version 2.0

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Revision History

Name	Date	Reason For Changes	Version
Siya Rath	15/4	System Overview Diagram Update + Updated ER Diagram	2
Aditi Sinha Siya Rath	18/4	Relational Schema Update + Normalization to 1NF	2
Asmita Guha	20/4	User Classes and Characteristics Update	2
Siya Rath Aditi Sinha Asmita Guha	22/4	Standard UI Components Finalization	2

1. Introduction

1.1 Purpose

FilmTrove is a unique **Movie Database Management System** carefully designed to store, manage and offer specific and organized information about movies, users, ratings, watch history, cast, genres, languages, and even streaming services. The main goal of FilmTrove is to **recommend movies** to users based on various features with the aim of increasing user interest with a more personalized approach.

This document outlines the **non-functional and functional requirements** of FilmTrove. It integrates all aspects of the system, including how a user manages accounts and records as well as administrative level controls. This document will be used by **developers, stakeholders, and testers** - any changes after the modifiable revision will be changed and noted on this document.

1.2 Document Conventions

Font and Text Formatting:

- Times New Roman is used throughout the document.
- Font size 12 for body text, font size 14 for side headings, and font size 18 for the title.
- Text is justified for a clean, uniform presence. Points are used whenever necessary.

Text Formatting:

- Bold highlight important terms.

Prioritization of Requirements:

- High-level requirements inherit the priority of detailed requirements unless stated otherwise.
- Each detailed requirement has its own priority.

1.3 Intended Audience and Reading Suggestions

The **Software Requirements Specification (SRS) for FilmTrove** is intended for the following stakeholders:

Developers

- Their focus is on the technical architecture, the data flow, and the details of how the system will be implemented.
- Key areas include user interface development, backend logic, database design, API security, and other integrative functions.

Project Managers

- They manage the schedules, milestones, and feature sets of a project.
- Ensure Scope: All functional, as well as non-functional, requirements should be addressed.

Quality Assurance (QA) Testers

- Create test routines and verification plans as per the requirements specified in the function and the system's intended performance.
- Key areas include system reactions, security breaches, and performance standards compliance.

End Users (Admins and Regular Users)

- **Admins:** Grasp the necessary knowledge about and experience in content moderation/review, approval workflow, and basic database administration.
- **Users:** Understand how FilmTrove assists users in obtaining, reviewing, and watching movies from the users' watchlists.

Marketing Staff

- Distill the primary characteristics of FilmTrove, as well as the associated advantages for users, that will be captured in the **marketing blurb**.
- Learn about how to escalate user retention campaigns, sources of differentiation, and the platform's distinctive features.

1.4 Product Scope

FilmTrove will offer users a seamless platform to **browse about movies, rate them, track their watched history and access metadata** related to movies such as cast, genre, and available streaming services. The primary focus of the system is to improve movie discovery and user interaction by providing **automated recommendations** based on watch history and ratings.

Notable features of FilmTrove are:

- Registration, login, and profile management.
- A comprehensive movie database containing information about the actors, director, and genres.
- Ratings for movies including reviews.
- History of watching movies for tailored recommendations.

- Integrated streaming services for their availability.
- Subtitles and dubbing information for other languages.
- Tailored movie suggestions that allow better experience for the user.

1.5 References

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- TMDb, *The Movie Database API*, Available: <https://www.themoviedb.org>
- Scribd, *Document Sharing and Publishing Platform*, Available: <https://www.scribd.com>

2. Overall Description

2.1 Product Perspective

- FilmTrove is a **standalone movie recommendation and database management system**, designed to provide users with **personalized movie recommendations** and structured access to movie-related data. It is **not a replacement for an existing system** but aims to enhance movie discovery by integrating user preferences, watch history, and ratings.
- FilmTrove includes several key components:
- **User Interface (UI)**: Enables users to browse, search, and receive recommendations.
- **Recommendation Engine**: Uses user preferences and collaborative filtering for personalized suggestions.
- **Movie Database Management**: Stores information about movies, genres, ratings, watch history, and streaming availability.
- **User Review System**: Allows users to rate and review movies.
- **Backend Database**: Manages and processes data efficiently for fast retrieval and accurate recommendations.
- FilmTrove is designed to integrate with **third-party movie databases (e.g., IMDb, TMDb)** and **streaming platforms** to provide comprehensive movie metadata. The system enhances the **movie discovery experience** by tailoring recommendations based on user preferences and viewing history.

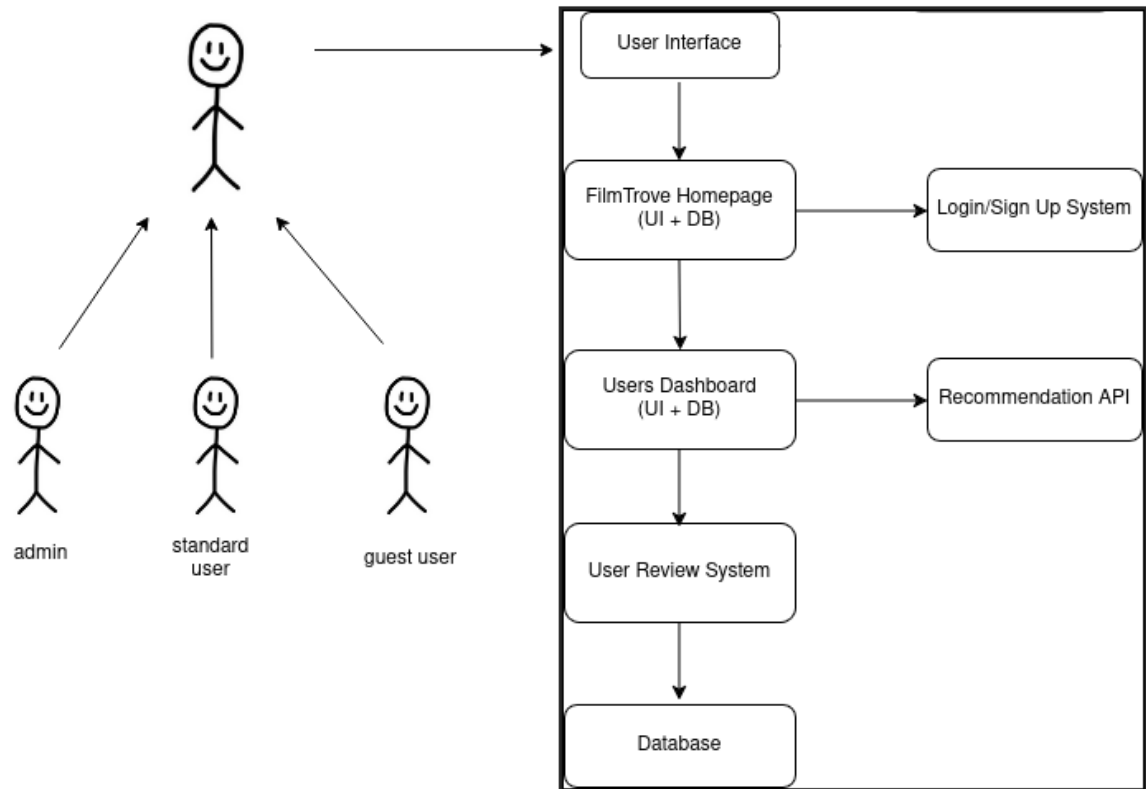


Fig: System Overview

2.2 Product Functions

Major functionalities of FilmTrove:

User Management:

- Register, log in, and manage user profiles.
- Store and update user preferences (preferred genres, languages, etc.).

Movie Database Management:

- Store movie details, including cast, director, genres, and ratings.
- Provide availability details for movies on various streaming platforms.

Movie Recommendations:

- Personalized suggestions based on **user preferences** and **watch history**.
- **Genre-based, collaborative filtering, and popularity-based recommendations.**

Watch History Tracking:

- Record movies a user has watched.
- Prevent recommending already watched movies.

Rating and Review System:

- Allow users to **rate and review movies**.
- Rank movies based on ratings and popularity.

Search and Filtering:

- Search movies by title, genre, language, or streaming service.
- Filter movies based on rating, and user preferences.

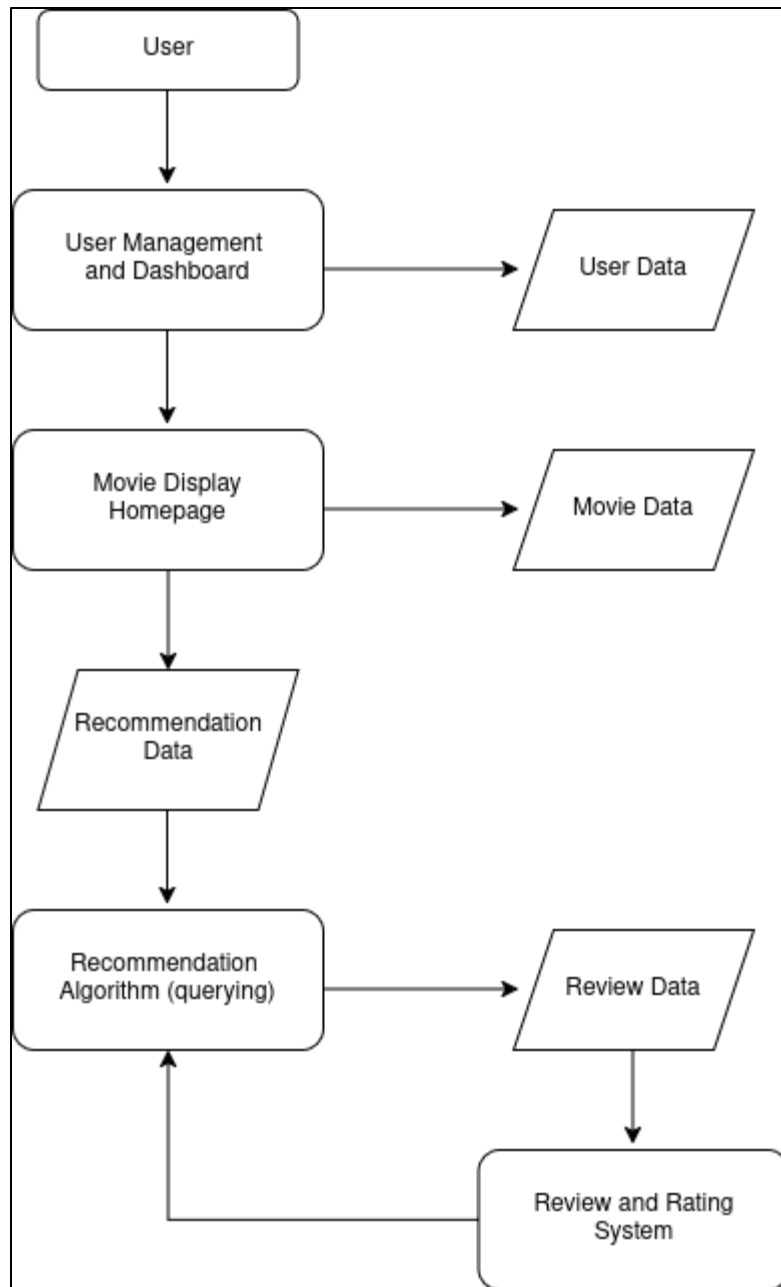


Fig: Top Level Data Flow Diagram

2.3 User Classes and Characteristics

Standard Users (Movie Enthusiasts)

- **Frequency of Use:** High

- **Functions:** Movie browsing, receiving recommendations, rating movies, managing watch history.
- **Technical Expertise:** Basic (no technical knowledge required).
- **Characteristics:** Users seek **easy-to-use navigation**, quick access to movie details, and personalized suggestions.

Admin Users

- **Frequency of Use:** Moderate to High
- **Functions:** Managing user accounts, updating movie database, moderating reviews.
- **Technical Expertise:** Intermediate (knowledge of backend and data management).
- **Characteristics:** Admins ensure smooth platform operation and data integrity.

Guest Users

- **Frequency of Use:** Low to Moderate
- **Functions:** Browsing movie details (without personalized recommendations).
- **Technical Expertise:** Basic
- **Characteristics:** Encouraged to register for enhanced features.

The most critical users of **FilmTrove** are **Regular Users (Movie Enthusiasts)** and **Admin Users**, as they directly impact the platform's success and functionality. Regular Users drive engagement by **browsing movies, rating films, and managing their watch history**, making them the primary audience. Their interactions shape recommendations and influence content visibility. Admin Users ensure **smooth platform operations**, managing user accounts, moderating reviews, and keeping the movie database up to date. Without these two user groups, the platform would lack both active participation and proper content management, making them essential for **FilmTrove's overall functionality and user experience**.

2.4 Operating Environment

FilmTrove is a **web-based application**, requiring various components to work seamlessly:

Hardware Requirements:

- **Processor:** 2.5 GHz multi-core CPU
- **Memory:** 8 GB RAM
- **Storage:** 500 GB SSD

- **Network:** High-speed internet connection

Operating System:

- **Development:** Windows 10/11 (XAMPP for local hosting)
- **Production:** Linux-based servers (e.g., Ubuntu 20.04+, CentOS) for cloud deployment on AWS/DigitalOcean
- **Cross-Platform Support:** macOS 11+ for development and testing

Software Components:

1. Backend & Database

- **C# (.NET Core):** Manages business logic, processes user requests, and handles backend operations.
- **SQLPlus 8.0+:** Executes queries to generate **movie recommendations** without a dedicated recommendation engine.

2.1.1.1 Frontend Technologies

- **C# (.NET with Blazor/WPF/WinForms as applicable):** Provides an interactive FilmTrove UI for a smooth user experience.
- **HTML5, CSS3, JavaScript, Bootstrap:** Enhances design and responsiveness.

2.1.1.2 Development Tools

- **Visual Studio:** Primary IDE for both frontend and backend development.
- **GitHub:** Used for version control and collaboration.
- **XAMPP:** Provides a local server environment for **development and testing**.

2.5 Design and Implementation Constraints

Corporate/Regulatory Policies

- FilmTrove must comply with **GDPR** and other data protection regulations to ensure user privacy and data security.

- Adherence to copyright laws and content licensing agreements is required for displaying movie metadata and linking to streaming platforms.

Database & API Constraints

- FilmTrove uses a **relational database model** designed based on ER diagrams.
- All movie metadata is sourced from **external APIs** such as IMDb and TMDb for accurate and up-to-date information.

Security & Authentication

- User authentication is implemented via **OAuth or JWT tokens** for secure login sessions.
- Sensitive user data (e.g., passwords, payment details) is encrypted using **SSL/TLS protocols** for secure transmission.

Performance Constraints

- Movie recommendations must be generated and displayed within **2 seconds** to maintain a seamless user experience.
- The system is designed to support **thousands of concurrent users** without performance degradation.

Development Technologies

- **Backend:** Development is restricted to **PHP and Python** for processing business logic and API interactions.
 - **Frontend:** The user interface must be built using **React.js or Angular** to ensure responsiveness and interactivity.
-

2.6 User Documentation

The following documentation will be provided:

- **User Manual:** Step-by-step guide for navigating the platform (PDF format).

- **Admin Guide:** Instructions on managing users and the movie database (PDF/HTML).
- **API Documentation:** Reference guide for developers integrating FilmTrove (Web-based).
- **FAQs:** Common user queries and troubleshooting solutions (Web-based).
- **Tutorials:** Interactive guides or videos on using the platform (MP4/HTML).
- **Release Notes:** Updates on new features, bug fixes, and improvements (Text/PDF).

2.7 Assumptions and Dependencies

Assumptions

- **Internet Access Required** – Users must have a stable internet connection to access FilmTrove’s features, including movie recommendations, watch history, and streaming service redirection.
- **External APIs Availability** – The system assumes that IMDb/TMDb APIs are consistently available for fetching movie metadata, ratings, and reviews.
- **Streaming Service Integration** – Valid API keys are required to connect to streaming platforms for seamless redirection to movie-watching options.
- **Scalability Considered** – FilmTrove is designed to efficiently handle high traffic, assuming the infrastructure can scale as needed.
- **User Familiarity with Web Navigation** – Users are expected to have basic experience navigating websites, requiring minimal training or onboarding.

Dependencies

- **Database and Query Execution** – FilmTrove relies on SQLPlus for executing queries and managing recommendations. The stability and efficiency of SQLPlus directly impact system performance.
- **Hosting and Infrastructure** – The availability and reliability of cloud hosting providers (AWS, DigitalOcean, or self-hosted Linux servers) affect system uptime and responsiveness.
- **Browser Compatibility** – The platform assumes full support for modern web technologies, including HTML5, CSS3, JavaScript, and C#-based Blazor components, across major web browsers (Chrome, Firefox, Safari, Edge).
- **Security Compliance** – FilmTrove must adhere to industry standards for data protection, ensuring encryption of all sensitive data and secure authentication protocols.
- **Legal and Regulatory Compliance** – The system assumes adherence to copyright laws and content distribution regulations regarding movie data usage, streaming links, and user-generated content.

3. External Interface Requirements

3.1 User Interfaces

FilmTrove offers a **responsive, web-based interface** designed using **C#, HTML5, CSS3, JavaScript, and Bootstrap**, ensuring an intuitive and engaging user experience across multiple devices. The frontend is developed and managed using **Visual Studio**.

3.1.1 Key UI Components

1. Homepage

- Welcome page
- Shows Logo and links to Login page
- Also links to a guest user login

2. Login Page

- Login using username and password
- Linked to Signup page

3. Signup Page

- Allows new users to create an account using **email and password**
- Captures **basic user details**, such as name and birth date

4. User Dashboard Page

- Displays **basic user details**
- **Has option for user to view Watch History and Ratings**
- **Personalized recommendations** will be provided (based on previous ratings and genres).

6. Watch History, Reviews and Ratings Page

- Displays a **comprehensive list of all watched movies**, including dates and duration

- Also displays review and rating of the user
- Updates the watch history accordingly as review is added

7. Recommendations Page

- Recommended movies based on Recommendation logic queries
- Displays Streaming platform for recommended movies
- Displays both personalized and filtered recommendations according to user requirements

8. Admin Access & Account Management Page

- Admin has both Movie Controls and User Controls
- Movie Controls allows admin to view all movies or insert another based on the inputs given
- User Controls allows admin to view all accounts signed into the system and delete any account as required for moderation

9. Reset Password Page

- Reset password functionality with old password and new password
- Security Question

3.1.2 Standard UI Components

Navigation Bar: Includes **Proceed, Back, Watch History, Get Recommendations** buttons that navigate between different forms (for authorized users).

Search and Filter Options: Users can search by **rating, genre, language, or duration** and filter results accordingly.

Common Buttons: "Search", "Add to Watchlist", "Login", "Sign Up" & "Submit Review".

Error Messages: Displayed as **message box pop-ups** with **tooltips** for additional guidance.

Keyboard Shortcuts: **Ctrl + F** → Focus on the search bar & **Esc** → Close pop-ups.

3.1.3 Software Components Requiring UI

User-Facing Modules

- a. Search
- b. Recommendations (via SQLPlus Queries)
- c. Reviews & Ratings
- d. Watch History Management

Admin Panel Modules

- a. Movie Management
- b. Account Moderation
- c. User Analytics & Logs

3.2 Hardware Interfaces

3.2.1 Supported Hardware Devices

3.1.1.1.1 **Web Servers:** Hosts the backend application (**C#/.NET Core**) and serves dynamic content over **HTTP/HTTPS**. Ensures **scalability** to support high user traffic and concurrent sessions.

3.1.1.1.2 **Database Server:** Runs **MySQL 8.0+** to store and manage movie metadata, user profiles, reviews, and watch history. Processes **SQLPlus queries** for generating recommendations, without a dedicated recommendation engine. Ensures **data integrity and optimized query performance** for smooth user experience.

3.1.1.1.3 **User Devices:** Accessible on **desktops, laptops, tablets, and smartphones** via major web browsers (**Chrome, Firefox, Safari, Edge**). **Responsive UI** designed for optimal viewing across different screen sizes.

3.1.1.1.4 **Streaming Devices (Future Expansion):** Potential **smart TV integration** for a seamless watchlist experience. Support for **casting movies to external screens** via compatible streaming devices.

3.2.2 Communication Protocols

3.1.1.1.5 **HTTP/HTTPS:** Secure data exchange between **web server, database, and user devices**. **SSL/TLS encryption** ensures user data privacy.

3.1.1.1.6 **SQL Queries: Backend (C#/.NET)** interacts with **MySQL via SQLPlus** to store and retrieve movie metadata, reviews, and recommendations. Optimized SQL queries for **fast response times**.

3.1.1.1.7 **WebSockets (Future Expansion):** Enables **real-time updates** on trending movies, new releases, and user interactions. Facilitates **live watch parties and shared viewing experiences**.

3.3 Software Interfaces

3.3.1 Operating System Compatibility

- Windows 10/11
- macOS 11+
- Linux-based servers (e.g., Ubuntu, CentOS)

3.3.2 Backend and Database

- **C# (.NET Core):** Handles business logic, processes user requests, and manages backend operations.
- **SQLPlus:** Executes queries to retrieve and generate movie recommendations without using a recommendation engine.
- **MySQL 8.0+:** Stores movies, user profiles, watch history, ratings, and reviews.

3.3.3 Frontend Technologies

- **C# (.NET with Blazor/WPF/WinForms as applicable):** Powers the FilmTrove UI for an interactive user experience.

3.3.4 Development Tools

- **Visual Studio:** Primary IDE for both frontend and backend development.
- **GitHub:** Version control and collaboration.
- **XAMPP:** Local server environment for development and testing on a larger scale.

3.3.5 Data Flow

- **Incoming Data:**
 - User inputs such as search queries, movie ratings, watchlist additions, and reviews.
- **Outgoing Data:**
 - Movie search results, recommendations (generated via SQLPlus queries), user feedback, and admin alerts.

3.3.6 Communication and Data Sharing

- **Frontend-Backend Communication:**
 - Uses RESTful APIs over HTTP/HTTPS for secure data exchange.
- **Database Interaction:**
 - Backend interacts with MySQL using SQLPlus queries for recommendations, retrieving user watchlists, and processing ratings.
- **Shared Data:**
 - User profiles, watchlists, movie metadata, ratings, and reviews are exchanged between the frontend, backend, and database.

3.3.7 Implementation Constraints

- All data transfers must be encrypted using HTTPS.
- SQLPlus queries will be used instead of a separate recommendation engine.

3.4 Communications Interfaces

FilmTrove requires **secure and efficient communication channels** for **user interactions, data retrieval, and system integration**.

3.4.1 Web Communication

- **HTTP/HTTPS** → Ensures **secure web browsing** and **encrypted data exchange** between users and the server.
- **RESTful APIs** → Used for **communication between the frontend (C# application) and the backend**, facilitating **search queries, movie details retrieval, and user interactions**.
- **SQLPlus Queries** → Used to **generate movie recommendations** based on user watch history and ratings.
- **WebSockets (Future Expansion)** → Enables **real-time updates**, such as **newly trending movies, live watch parties, and instant notifications for reviews or interactions**.

3.4.1 File Management & Notifications

- **FTP (Admin File Uploads & Management)** → Admins can upload **movie posters, trailers, and promotional content** securely.
- **SMTP/IMAP (User Email Notifications)** →
 - Sends **alerts for new recommendations, watchlist updates, and review responses**.
 - Notifies **admins about new reviews and content moderation tasks**.

3.4.2 Streaming & External Integrations (Future Scope)

- **Integration with Netflix, Prime Video, Disney+ APIs** → Fetches **streaming availability and direct watch links** for listed movies.

3.4.3 Security & Data Protection

- **TLS 1.2+ Encryption** → Ensures **secure data transfer** between all system components.
- **OAuth Authentication** → Secure **user login and session management**.

4. System Features

4.1 Movie Search and Filtering

4.1.1 Description and Priority

This feature allows users to search for movies and filter results based on various criteria such as genre, release year, language, and streaming platform availability.

- **Priority:** High
- **Benefit:** Enhances user experience by making it easier to discover movies.
- **Risk:** Low, as it is a core function of the platform.

4.1.2 Stimulus/Response Sequences

- **User Action:** The user enters a search query and selects filters (e.g., genre, year, streaming service).
- **System Response:** The system retrieves matching movies from the database using SQLPlus queries and displays results with relevant details (title, poster, description, rating, and streaming options).

4.1.3 Functional Requirements

- **REQ-1:** The system must allow users to search for movies using a keyword-based search.
- **REQ-2:** The system must support filtering by genre, release year, language, and streaming platform.
- **REQ-3:** If no results are found, the system must display an appropriate message: "No movies match your search criteria."

4.2 Movie Recommendations

4.2.1 Description and Priority

This feature provides users with movie recommendations based on their watch history and ratings using SQLPlus queries.

- **Priority:** High
- **Benefit:** Enhances user engagement by offering personalized suggestions.
- **Risk:** Medium, as recommendations depend on accurate data processing.

4.2.2 Stimulus/Response Sequences

- **User Action:** The user views the "Recommended Movies" section.
- **System Response:** The system retrieves recommendations based on the user's ratings and watch history, displaying them on the homepage.

4.2.3 Functional Requirements

- **REQ-7:** The system must generate movie recommendations using SQLPlus queries instead of a recommendation engine.
- **REQ-8:** Recommendations must be updated when a user adds new ratings or watchlist entries.
- **REQ-9:** If no recommendations are available, display the message: "Start rating movies to get personalized recommendations!"

4.3 User Reviews and Ratings

4.3.1 Description and Priority

Users can rate movies and submit reviews, contributing to the platform's movie database.

- **Priority:** Medium
- **Benefit:** Enhances platform content and helps other users make informed choices.
- **Risk:** Low, but improper moderation may affect platform credibility.

4.3.2 Stimulus/Response Sequences

- **User Action:** The user selects a movie and submits a rating (1-5 stars) with an optional review.
- **System Response:** The system saves the review and updates the movie's average rating.

4.3.3 Functional Requirements

- **REQ-10:** Users must be able to rate a movie and submit a review.
- **REQ-11:** The system must store and display reviews under the respective movie details page.
- **REQ-12:** Admins must be able to approve, delete, or flag inappropriate reviews.

4.4 Admin Panel for Movie and User Management

4.4.1 Description and Priority

This feature enables administrators to manage movies, user accounts, and reviews.

- **Priority:** High
- **Benefit:** Essential for maintaining platform integrity and data accuracy.
- **Risk:** Medium, as improper management could disrupt operations.

4.4.2 Stimulus/Response Sequences

- **User Action:** The admin logs into the panel and selects a movie, user, or review to manage.
- **System Response:** The system displays management options such as edit, delete, or approve.

4.4.3 Functional Requirements

- **REQ-13:** Admins must be able to add, edit, and remove movies from the database.
- **REQ-14:** Admins must be able to approve, delete, or flag reviews.
- **REQ-15:** Admins must have access to user activity logs for moderation purposes.

4.5 User Authentication and Security

4.5.1 Description and Priority

This feature ensures secure user authentication using OAuth or JWT-based login.

- **Priority:** High
- **Benefit:** Protects user data and ensures authorized access.
- **Risk:** High, as security breaches could compromise user privacy.

4.5.2 Stimulus/Response Sequences

- **User Action:** A user enters login credentials and clicks "Sign In."
- **System Response:** The system verifies credentials and grants access to the platform.

4.5.3 Functional Requirements

- **REQ-16:** Users must log in using OAuth authentication (Google, Facebook, etc.) or a standard username/password.
- **REQ-17:** Passwords must be encrypted before storage.
- **REQ-18:** If login fails, the system must display: "Invalid credentials. Please try again."

4.6 Streaming Service Integration (*Future Expansion*)

4.6.1 Description and Priority

This feature enables integration with streaming services like Netflix, Prime Video, and Disney+ to fetch movie availability.

- **Priority:** Medium
- **Benefit:** Allows users to directly access streaming platforms from FilmTrove.
- **Risk:** Medium, as it depends on external API availability.

4.6.2 Stimulus/Response Sequences

- **User Action:** The user clicks "Watch Now" on a movie.
- **System Response:** The system retrieves streaming availability via external APIs and redirects the user.

4.6.3 Functional Requirements

- **REQ-19:** The system must retrieve streaming availability from third-party APIs.
- **REQ-20:** If a movie is unavailable, display: "Currently not available on streaming platforms."

4.8 Email and Notification System (*Future Expansion*)

4.8.1 Description and Priority

Users receive email notifications for important updates like watchlist additions, recommendations, and account security alerts.

- **Priority:** Medium
- **Benefit:** Keeps users engaged with platform updates.
- **Risk:** Low, but excessive notifications could annoy users.

4.8.2 Stimulus/Response Sequences

- **User Action:** A user adds a movie to the watchlist.
- **System Response:** The system sends an email notification confirming the addition.

4.8.3 Functional Requirements

- **REQ-21:** The system must send email notifications for account updates, watchlist additions, and admin alerts.
- **REQ-22:** Users must be able to customize notification preferences.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- System should be able to handle a **minimum of 5000 users** logged in simultaneously without any substantial decline in performance.

- Movie search and filtering queries should be executed within **2 seconds** even with a **maximum of 5 million records** in a single database table.
- Loading times for all important pages like homepage, movie details page, and search result pages are capped at **3 seconds** in an average network condition.
- Submission of reviews and updating the watchlist should be done within **5 seconds** after a user initiates the action.
- System availability must be **99.9% inclusive** of scheduled downtime which does not go over **2 hours** per month.
- **Third party streaming APIs** check for availability in seconds must not exceed the **5 second** mark for an ideal experience.

5.2 Safety Requirements

- All data pertaining to users, such as watchlists, ratings, and reviews, should have their **daily backups** protected from data loss.
- **Database API scrubbing filter** must be enabled to block users from **damaging data (SQL injection, XSS) manipulation**.
- **Unauthorized editing or deleting** of movies, reviews, and even admin settings should be role protected with an **RBAC security** model.
- All user data needs guaranteed protection in storage and processing in compliance with **GDPR** and other relevant laws by FilmTrove.
- Users must have the ability to **export and delete their personal data** upon request to comply with privacy regulations.
- If a movie is removed from the system, users must be notified and provided with alternatives (if applicable).

5.3 Security Requirements

- Login will be implemented using **OAuth 2.0 authentication** with Google, Facebook, or email login options.
- **Multi-factor authentication** will be mandatory for admins and optional for regular users.
- Users will be logged out after **15 minutes of inactivity** to prevent unauthorized access.
- All passwords must be hashed with **bcrypt or Argon2 encryption** and stored securely.
- Potential security threats and vulnerability audits will be conducted regularly.
- All communication between the users and server must have **data encryption (TLS 1.2+)** applied.
- All admin activities, including but not limited to movie database updates, review moderation, and others must be logged for auditing and reviewed periodically.
- Third-party services for streaming must secure API keys and authenticate using **OAuth for API access**.

5.4 Software Quality Attributes

- **Usability:** No additional software should be required to access the user interface from a desktop or mobile device, and it must be easy to navigate.
- **Reliability:** The service must be running 24/7 alongside the availability of the system. The system should undergo optimizations that require no more than **2 hours of downtime** in a month.
- **Scalability:** The system architecture should enable changes to be made in the future to accommodate as far as **50,000 users along with millions of records of movies** without the system performance declining.
- **Maintainability:** The codebase should be **modular and well-documented**, making it easy for developers to update and fix bugs.
- **Portability:** The system must function across **all modern web browsers (Chrome, Firefox, Edge, Safari) and mobile devices** without requiring additional installations.
- **Interoperability:** APIs should be designed to allow **seamless integration with third-party platforms**, including **streaming services and user authentication providers**.
- **Testability:** The system should support **automated testing** for major components, including movie search, reviews, recommendations, and user authentication.
- **Robustness:** The system should be able to **handle unexpected user inputs gracefully**, displaying meaningful error messages instead of crashing.

5.5 Business Rules

- **User Registration:** Users are required to register to receive personal recommendations, add movies to their watchlist, or submit reviews.
- **Review Moderation:** All reviews will undergo an automatic scanning process for moderation. All reviews will need an admin approval prior to becoming visible to any user.
- **Admin Privileges:** Admins hold unrestricted access to the movie database as well as reviews and system/user activity logs.
- **Streaming Availability:** Partners such as Netflix, Prime Video, and Disney+ are the only ones who will be displayed on the platform.
- **Account Security:** Before reviewing or rating movies, users must verify their email to account for security reasons.
- **Inactive Accounts:** A user account that is inactive for 12 months will be archived or deleted with a notification sent 30 days prior.
- **Refund & Cancellation Policy:** A refund policy of up to 7 days will be issued for subscription-based services in the case that a premium feature is added in the future.

6. Other Requirements

6.1 Database Requirements

- **Relational Database:** The system will store **movie details, user profiles, reviews, watchlists, and admin logs** in a **relational database** such as **MySQL, PostgreSQL, or MongoDB**.
- **Normalization:** The database must be **normalized to at least Third Normal Form (3NF)** to ensure data integrity and reduce redundancy.
- **ACID Compliance:** The database must support **ACID (Atomicity, Consistency, Isolation, Durability)** properties for reliable transactions.
- **Backup Strategy:** A **full database backup must be performed every 24 hours**, with incremental backups every 6 hours to minimize data loss.
- **Indexing & Performance Optimization:** Queries must be optimized through **proper indexing and caching strategies** to maintain fast response times.

6.2 Internationalization Requirements

- **Language Support:** FilmTrove will initially support **English**, with future expansion to **French, Spanish, and German**.
- **Date & Time Formatting:** Dates should be displayed in the user's preferred format (e.g., **MM/DD/YYYY for U.S. users, DD/MM/YYYY for European users**).
- **Currency Formatting:** If FilmTrove introduces premium services, **currency should be displayed based on the user's locale**.
- **Movie Titles & Descriptions:** If available, **localized movie titles and descriptions should be displayed based on user preferences**.

6.3 Legal and Compliance Requirements

- **GDPR & Data Protection Compliance:** FilmTrove must adhere to **GDPR, CCPA, and other relevant data protection laws**, ensuring secure handling of user data.
- **User Consent:** Before using the platform, users must **agree to the Privacy Policy and Terms of Service**.
- **Copyright Compliance:** Movie metadata, images, and streaming links must be sourced **legally from authorized APIs (e.g., TMDb, IMDb, or official streaming platforms)**.
- **Data Retention Policies:**

Inactive user accounts should be archived after **12 months** and deleted after **24 months** (with prior email notifications).

Users must be able to **request data deletion** in compliance with **right-to-be-forgotten regulations**.

6.4 Reuse Objectives

- **Modular Architecture:** The system should be built in a **modular way**, allowing **components like search, recommendations, and watchlists** to be reused in future projects.
- **Use of Open-Source Technologies:** FilmTrove should **leverage open-source libraries** (e.g., **React.js, Bootstrap, SQL-based solutions**) to **minimize costs** and improve maintainability.
- **Streaming API Integration:** The API integration for streaming platform availability should be designed for **easy adaptation** to additional platforms (e.g., Hulu, Apple TV+).

Appendix A: Glossary

- **SRS** – Software Requirements Specification
- **GDPR** – General Data Protection Regulation
- **CCPA** – California Consumer Privacy Act
- **ACID** – Atomicity, Consistency, Isolation, Durability (database properties)
- **RBAC** – Role-Based Access Control
- **3NF** – Third Normal Form (database normalization standard)
- **SQL Injection** – A security vulnerability that allows attackers to interfere with database queries
- **TMDb** – The Movie Database (a popular movie and TV metadata API)
- **OAuth 2.0** – A secure authentication framework for user logins

Appendix B: Analysis Models

- **High-Level System Architecture Diagram:**

Showcases **how different modules interact**, including **user authentication, search, recommendations, watchlists, and admin management**.

- **Entity-Relationship (ER) Diagram:**

Illustrates database entities such as **Users, Movies, Reviews, Watchlists, and Admin Logs**, along with their **relationships**.

Appendix C: To Be Determined List

1. **Final Database Technology:** Use of MySQL, PostgreSQL, or MongoDB
2. **Language Support:** Additional languages to be prioritized in the first phase of internationalization
3. **Hosting Environment:** To decide whether to be hosted on AWS, Google Cloud, or Microsoft Azure
4. **Third-Party API Integration:** Streaming APIs (e.g., Netflix, Prime Video, Disney+) to be officially integrated first
5. **Monetization Strategy:** Premium features (e.g., **ad-free experience, advanced recommendations**) to be introduced in future versions?

Chapter 4

Data Design

1.1 ER Diagram

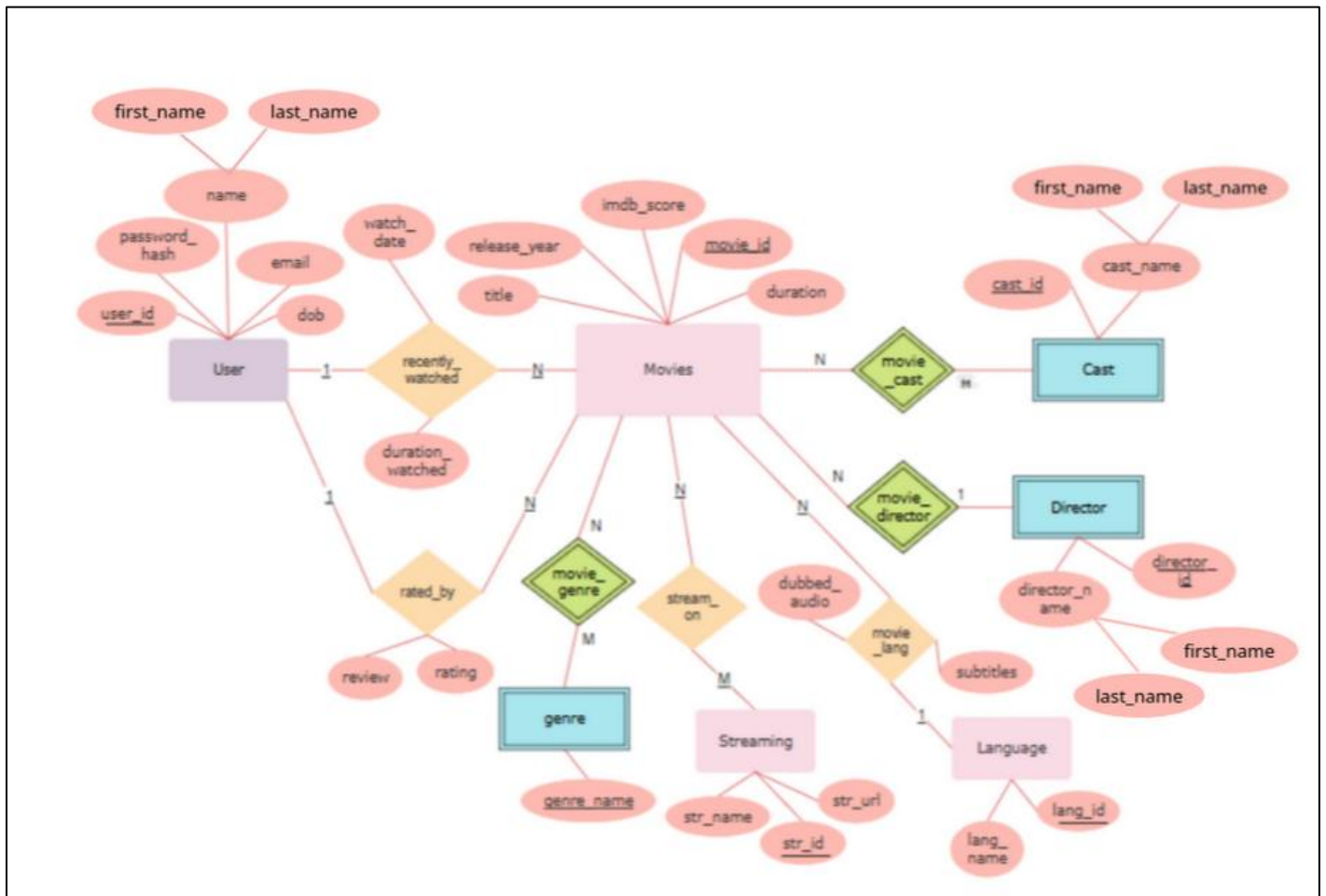


Figure 4.1: ER Diagram

1.2 Entity Description

The data design of FilmTrove encompasses core entities essential for managing users, movies, watch history, ratings, reviews, genres, streaming platforms, and language preferences. These entities form the foundation of a robust movie recommendation and management system, enabling smooth user interaction, content discovery, and personalized suggestions.

4.2.1 User

Represents individuals registered in the system, including movie enthusiasts, guest and admin.

Attributes:

- user_id: Unique identifier for each user (Primary Key).
- email: Unique email address.
- password_hash: Encrypted password for account access.
- first_name: first name of the user.
- last_name: last name of the user
- dob: Date of birth.

4.2.2 Movies

Stores the information about each movie and the details.

Attributes:

- movie_id: Unique movie identifier (Primary Key).
- title: Movie title.
- release_year: Year of release of the movie.
- duration: Length of movie in minutes.
- imdb_score: 3,2 decimal number rating given by user.
- director_first_name: first name of director.
- director_last_name: last name of director.

4.2.3 Cast

Lists information about the cast of the movies.

Attributes:

- cast_id: Unique identifier for each cast (Primary Key).
- cast_first_name: first name of cast member.
- cast_last_name: last name of cast member.

4.2.4 Movie Cast

Relates the cast of the movies to the movie itself to be added into the recommendation's algorithm.

Attributes:

- movie_id: Unique identifier for each movie (Primary Key).
- cast_id: Unique identifier for each cast member (Primary Key).

4.2.5 Streaming

Tracks which streaming service each movie is on.

Attributes:

- str_id: Unique identifier for each streaming platform (Primary Key)
- str_name: Name of the streaming website.
- str_url: URL under which you can access each website.

4.2.6 Movie_Streaming

Maintains information about which streaming service each movie is present on.

Attributes:

- movie_id: Unique identifier for each movie (Primary Key).
- str_id: Unique identifier for each streaming platform (Primary Key).

4.2.7 Language

Stores information about the language of the movie in question.

Attributes:

- lang_id: Unique identifier for each language (Primary Key).
- lang_name: Name of every unique language.

4.2.8 Movie_Language

Maintains information about which language each movie is present in.

Attributes:

- movie_id: Unique identifier for each movie (Primary Key).
- lang_id: Unique identifier for each unique language (Primary Key).
- dubbed_audio: boolean value represented by y/n
- subtitles: boolean value represented by y/n, tells user if subtitles are present

4.2.9 Rating

Rating and review system for tracking movies viewed by users.

Attributes:

- user_id: Unique identifier for each user (Primary Key).
- movie_id: Unique identifier for each movie (Primary Key).
- review: Short review entered by the user for each movie.
- rating: Rating given for each movie in 3,2 decimal numbers.

4.2.10 Recently Watched

Maintains information about movies recently watched by non-guest users.

Attributes:

- user_id: Unique identifier for each user (Primary Key).
- movie_id: Unique identifier for each movie (Primary Key).
- watch_date: Date on which the movie was logged by the user. (Primary Key)
- duration_watched: Length of movie watched by user.

4.2.11 Movie Genre

Lists out movies and their corresponding genres.

Attributes:

- movie_id: Unique identifier for each movie (Primary Key).
- genre_name: Genre of the movie as logged by the system. (Primary Key)

1.3 Normalization

1.3.1 First Normal Form (1NF)

In the original schema:

- The name attribute (representing a user's full name) was a **composite attribute**.
- The director_name attribute (representing the full name of a director) was also **non-atomic**.
- The cast_name attribute (representing the full name of a cast member) was also **non-atomic**.

To comply with 1NF:

- name is **decomposed into** first_name and last_name
- director_name is **decomposed into** director_first_name and director_last_name
- cast_name is **decomposed into** cast_first_name and cast_last_name

1.3.2 Second Normal Form (2NF)

Second Normal Form (2NF) To achieve 2NF, the table must first be in 1NF. Then, it must have no partial dependencies (i.e., no non-prime attribute should be dependent on a part of the primary key).

Applying 2NF to the schema:

- We observe partial dependency in movie_cast((movie_id, cast_id) → cast_name) where cast_name only depends on cast_id not movie_id. Hence, we create a separate cast relation.
- All the remaining tables are in 2NF.

1.3.3 Third Normal Form (3NF)

Third Normal Form aims to remove transitive dependencies, cases where a non-prime attribute depends on another non-prime attribute, rather than directly on the primary key.

All relations have no transitive dependency and satisfy the above conditions. Hence, R is in 3NF.

1.3.4 Boyce Codd Normal Form (BCNF)

A table is in BCNF if it is in 3NF and if every determinant is a superkey. All functional dependencies have only candidate keys as determinants. There are no non-trivial dependencies violating BCNF.

Since all dependencies involve primary keys and there are no partial or transitive dependencies left, the schema is now in BCNF.

1.3.5 Reduced and Normalized Schema:

1. **db_user**(user_id VARCHAR(10) primary key, first_name VARCHAR(255), last_name VARCHAR(255), email VARCHAR(255), password_hash VARCHAR(255), dob DATE)
2. **db_Movies**(movie_id VARCHAR(10) primary key, title VARCHAR(255), release_year INT, duration

INT, imdb_score NUMBER(3,2), director_first_name VARCHAR(255), director_last_name VARCHAR(255));

3. **dbb_Cast**(cast_id VARCHAR(10) primary key, cast_first_name VARCHAR(255), cast_last_name VARCHAR(255));
4. **dbb_Movie_Cast**(movie_id VARCHAR(10), cast_id VARCHAR(10), primary key(movie_id, cast_id));
5. **dbb_Streaming**(str_id VARCHAR(10) primary key, str_name VARCHAR(255), str_url VARCHAR(255));
6. **dbb_Movie_Streaming**(movie_id VARCHAR(10), str_id VARCHAR(10), primary key(movie_id, str_id));
7. **dbb_Language**(lang_id VARCHAR(10) primary key, lang_name VARCHAR(255));
8. **dbb_Movie_Language**(movie_id VARCHAR(10), lang_id VARCHAR(10), dubbed_audio BOOLEAN, subtitles BOOLEAN, primary key(movie_id, lang_id));
9. **dbb_Rating**(user_id VARCHAR(10), movie_id VARCHAR(10), rating NUMBER(3,2), review VARCHAR(255), primary key(user_id, movie_id));
10. **dbb_Recently_Watched**(user_id VARCHAR(10), movie_id VARCHAR(10), watch_date DATE, duration_watched INT, primary key(user_id, movie_id, watch_date));
11. **dbb_Movie_Genre**(movie_id VARCHAR(10), genre_name VARCHAR(255), primary key(genre_name, movie_id));

Chapter 5

Methodology

5.1 System Design Model

5.1.1. Homepage

The Homepage serves as the landing interface for all users—new, returning, or guests. This design ensures a welcoming experience redirects to the login page. Responsive layout, intuitive design, and visual branding make the homepage the gateway into FilmTrove’s recommendation ecosystem.

5.1.2. Login Module

The login module is the access checkpoint for authenticated features. Users log in using their registered email and password. The system checks for role-based access:

- **Admin:** Redirected to the admin access control panel.
- **Standard User:** Redirected to their user dashboard.
- **Invalid credentials** trigger immediate error messages via MessageBox, preventing unauthorized access. Input validation ensures email formatting and password field completion.

5.1.3. Signup Module

5.1.3. Signup Module

It collects:

- Full name (letters only),
- Unique email ID (validated using Regex),
- Secure password (7–77 characters),
- Confirm password,
- Birth date (used to calculate age),
- Security question/answer (for password recovery)

New users must be **18 years or older**, with real-time validation and feedback. Upon submission, the system generates a new unique user_id using the GenerateNextUserId() function, and the data is stored in dbs_User table.

5.1.4. Reset Password Module (Security Question Feature)

Users who forget their password can recover it using the **Reset Password** form. The user enters their email in the login form and is prompted with their associated security question in the reset password. If the answer matches, the user is allowed to set a new password. This secure recovery method adds an extra layer of protection, with validation at each step to ensure user identity.

5.1.5. Guest User Mode

Guest users can explore FilmTrove without an account. They are limited to the **Filtered Recommendations** feature, where they can apply filters like genre, language, or duration of movie. No watch history, ratings, or personalization features are available in guest mode, which maintains system security and data separation.

5.1.6. Admin User Module

Admin access provides a robust control interface for managing:

- **Movie Listings:** View and verify all movie records.
- **Add Movies:** Insert new records into `db_Movies`, including linking genre, streaming platforms, and languages.
- **User Controls:** View all registered users and delete users with inappropriate activity or invalid data.

The module incorporates internal ID generation (e.g., `GenerateNextMovieId()`), dropdowns populated from `db_Genre`, `db_Language`, and `db_StreamingPlatform`, and full input validation to prevent malformed entries. Admin dashboards also display real-time metrics for system activity and user base growth.

5.1.7. Movie Listing Page (Admin Only)

This admin-only interface displays all movies stored in the database, including movie ID, title, language etc.,. It is used for administrative auditing and allows verification and deletion if inconsistencies arise.

5.1.8. User Dashboard (Standard Users)

After login, users land on a dashboard displaying:

- Personal details (name, email, DOB)
- Navigation to Watch History
- Navigation to Recommendations

The dashboard provides a personalized view, enhancing user retention and interaction. It is secured via session-based role authentication.

5.1.9. Watch History Module

Users' viewing activity is tracked in the `db_RecentlyWatched` table. This page displays:

- Previously watched movies (titles, watched date, rating, review)
- Options to rate/review newly watched movies. Each review entry updates `db_Rating`, which is then used for generating recommendations. The page promotes interaction and community-driven content quality.

5.1.10. Recommendations Module

FilmTrove's recommendation engine operates in two modes:

- **Personalized Recommendations:**
 - Based on genres and ratings from watch history
 - Custom SQL queries run against `db_RecentlyWatched`, `db_Movie_Genre`, and `db_Rating`
 - Tailored suggestions unique to each user

- **Filtered Recommendations:**
 - Open to both registered and guest users
 - Filters include genre, language, platform, release year, and IMDb score.
 - Provides instant movie discovery via live query generation.

Both modes use pre-defined and dynamic PL/SQL procedures and ensure fast, relevant results.

5.2 Validation and Error Handling

FilmTrove uses **multi-layer validation**:

- **Client-side validation:**
 - Regex on name, email, password match
 - Drop-downs prevent null selection
 - Date of birth and age logic for age eligibility
- **Server-side checks:**
 - Unique ID generation using auto-increment logic
 - Check for existing emails to prevent duplicates
 - Secure command parameterization to avoid SQL injection
- **Error Handling:**
 - Try-catch blocks in all DB operations
 - MessageBox.Show() alerts for invalid input, failed connections, or mismatched passwords
 - Error messages are user-friendly, guiding toward resolution

5.3 System Navigation and Flowcharts

FilmTrove is structured around clearly defined roles—**Admins**, **Registered Users**, and **Guests**—each accessing a customized interface tailored to their needs. The navigation and interaction flow between users and system modules are illustrated through dedicated block diagrams.

5.3.1. Admin Access Flow

Administrators manage the backend of the FilmTrove platform. After logging in via a secure form, they are redirected to a dashboard with comprehensive control capabilities, including:

- **Insert Movies:** Add new movie data, including genre, language, platform, and associated metadata.
- **Movie Listing (View Only):** Access a complete list of movies in the database.
- **User List (View/Delete):** Monitor all users and remove users if necessary.
- **Input Validation:** Error handling with messages.

All actions are completely validated before data is committed to the respective database tables, ensuring consistency and security.

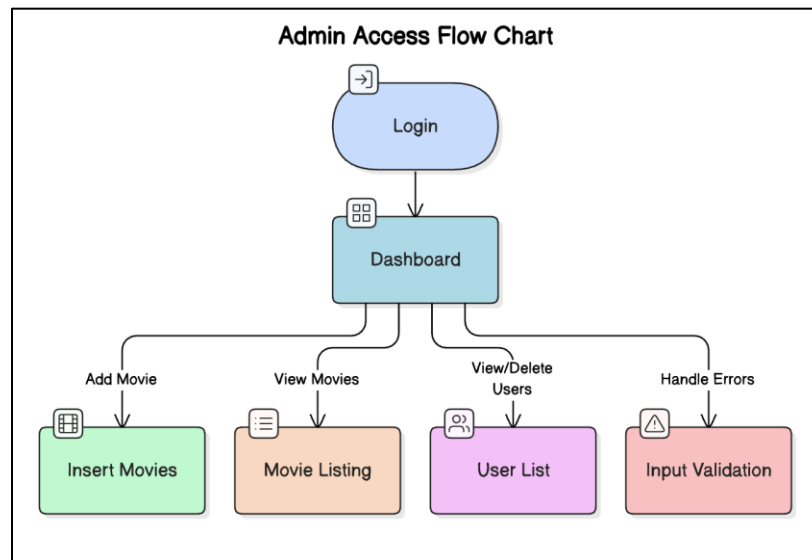


Figure 5.1: Admin Access Flow Chart

5.3.2. Registered Users Login and Navigation Flow

Standard users gain access to FilmTrove through a login interface. Once authenticated, they are redirected to their **User Dashboard**, which includes:

- **Personal Profile:** View personal data such as name, email, and date of birth.
- **Watch History:** View all previously watched films. Users can rate and review new movies, which are automatically added to the history.
- **Recommendations Page:**
 - **Filtered Recommendations:** Users apply filters (e.g., duration, IMDb rating, genre, language) to explore new movies.
 - **Personalized Recommendations:** Based on watch history, genre preferences, and ratings—tailored via complex SQL-based filtering queries.

All user interactions are protected with robust input validation and error feedback mechanisms.

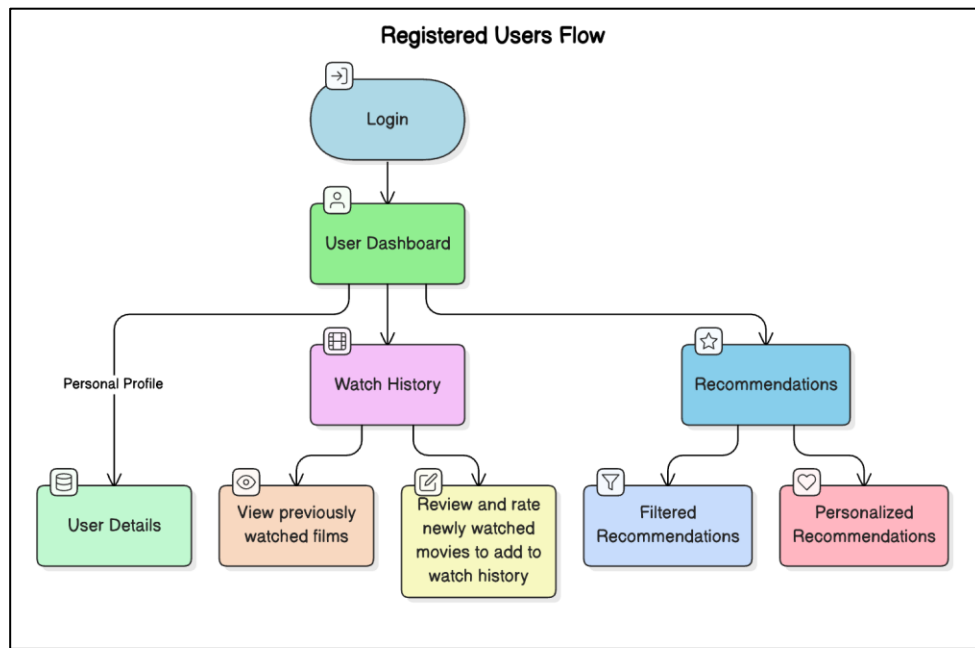


Figure 5.2: Registered Users Flow

5.3.3. Guest User Flow

Guest users are granted **limited access** to explore movies through the **Filtered Recommendations** module. This mode does not require login or registration and ensures that new users can still experience the core discovery features without contributing data or receiving personalization.

- **Read-only Access**
- **No watch history or review options**
- **Encourages registration by offering a taste of the platform**

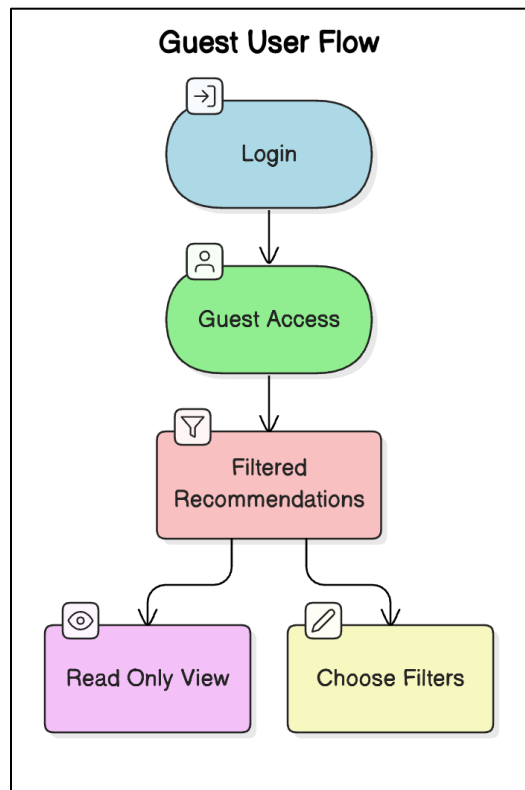


Figure 5.3: Guest User Flow

5.3.4. Movie Data Lifecycle Flow

Admin Adds Movie → Inserts into:

- dbs_Movies
- dbs_Movie_Genre
- dbs_Movie_Language
- dbs_Movie_Streaming

↓

Users View and Interact

- Watch via external streaming platforms (linked)
- Submit Ratings and Reviews (stored in dbs_Rating)
- Recently watched stored in dbs_RecentlyWatched
- Recommendations updated

↓

Guest Users

- Access only public movie filters (read-only, non-personalized)

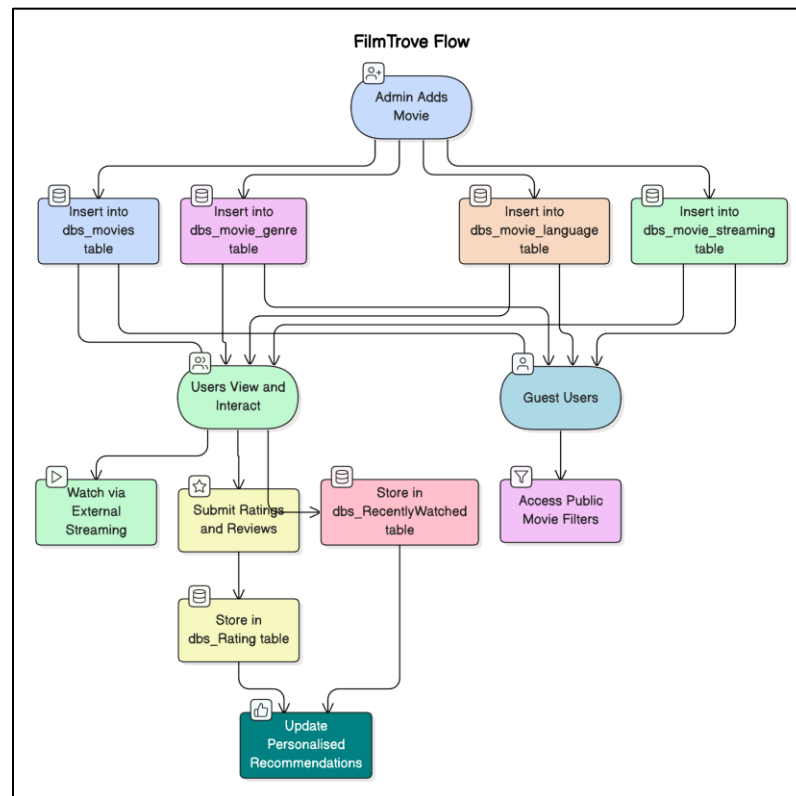


Figure 5.4: Movie Data Flow Diagram

Summary of Roles and Functionalities

Role	Accessible Modules	Permissions
Admin	Dashboard, Movie Control, User Control	Full CRUD operations on data
Standard User	Dashboard, Watch History, Recommendations	View, Rate, Review, Filter
Guest User	Filtered Recommendations	Read-only browsing

5.4 Development Tools

Component	Tool/Tech
Frontend	Visual Studio – C# (Windows Forms), Canva
Backend	Oracle SQLPlus (PL/SQL & SQL), OracleCommand in C#
Database	BCNF-normalized relational schema with tables like Users, Movies, Cast, Genres, Streaming Platforms, Languages, Ratings, Watch History
Validation	Regex, C# logical conditions, error messages via MessageBox
Security	Role-based login, secure password checks, security question for password recovery
Recommendation Logic	SQL-based query filtering using multiple table joins and user behavior analysis

Chapter 6

Results

6.1 Home Page

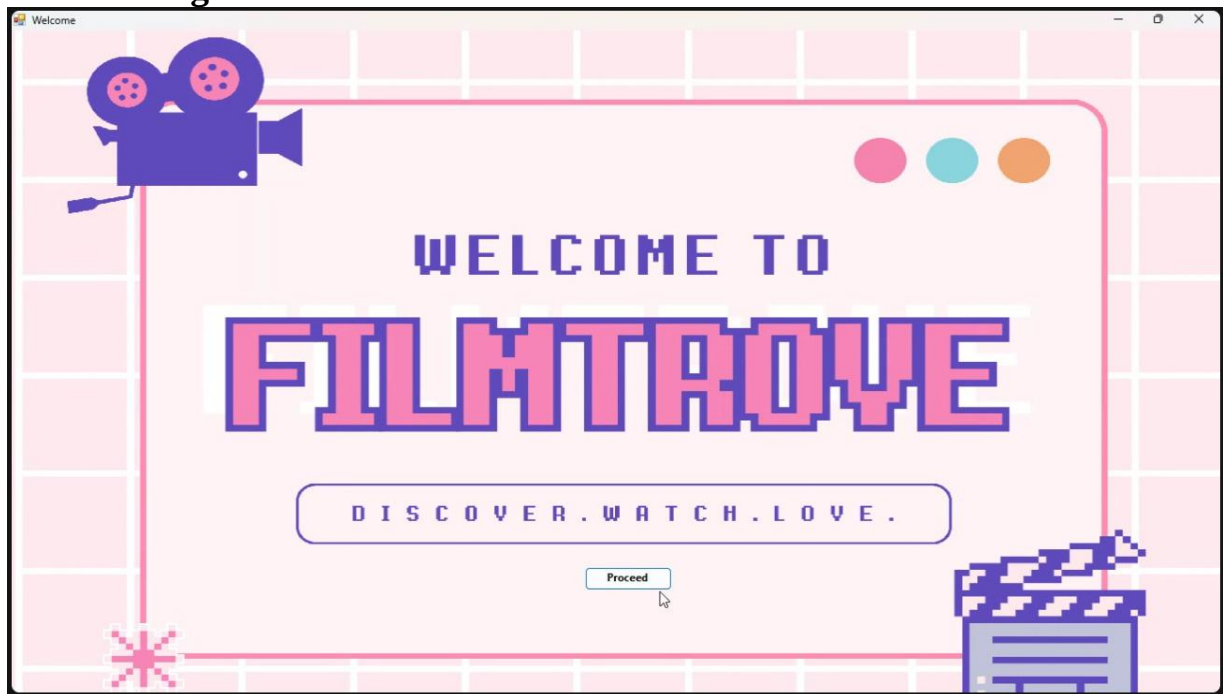


Figure 6.1: Home Page

6.2 Login Page

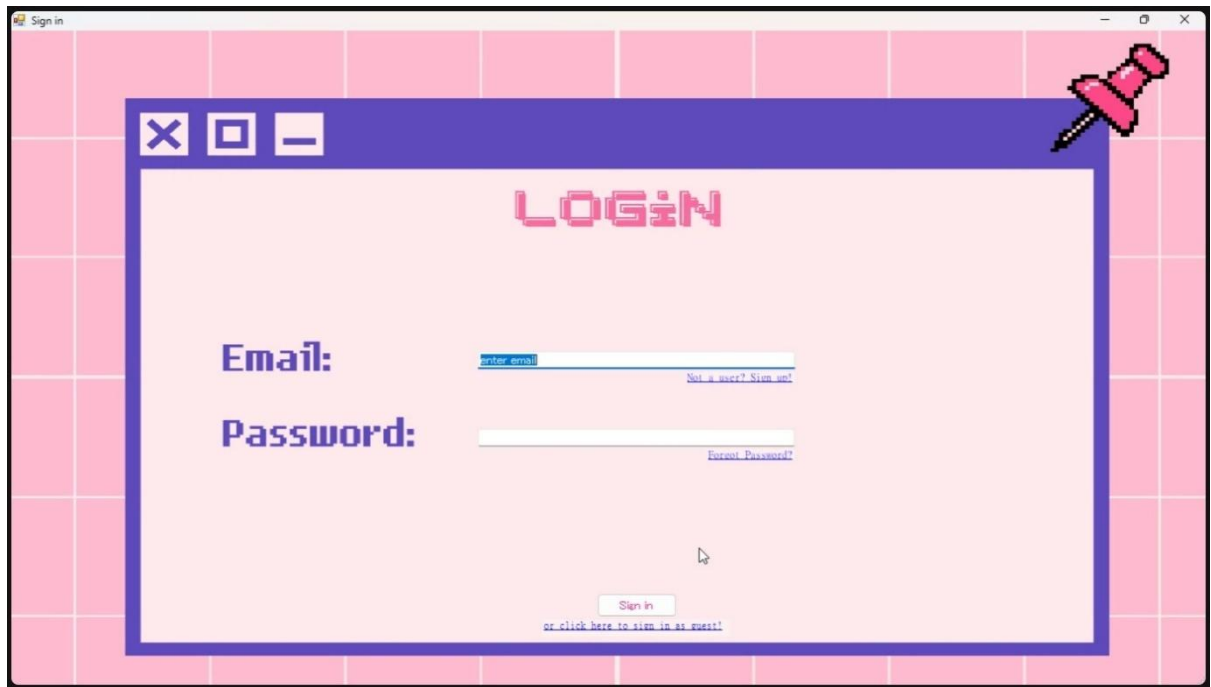


Figure 6.2: Login Page with options for Standard Login, Guest Login, Sign Up and Forgot Password

6.2.1 Standard User Login

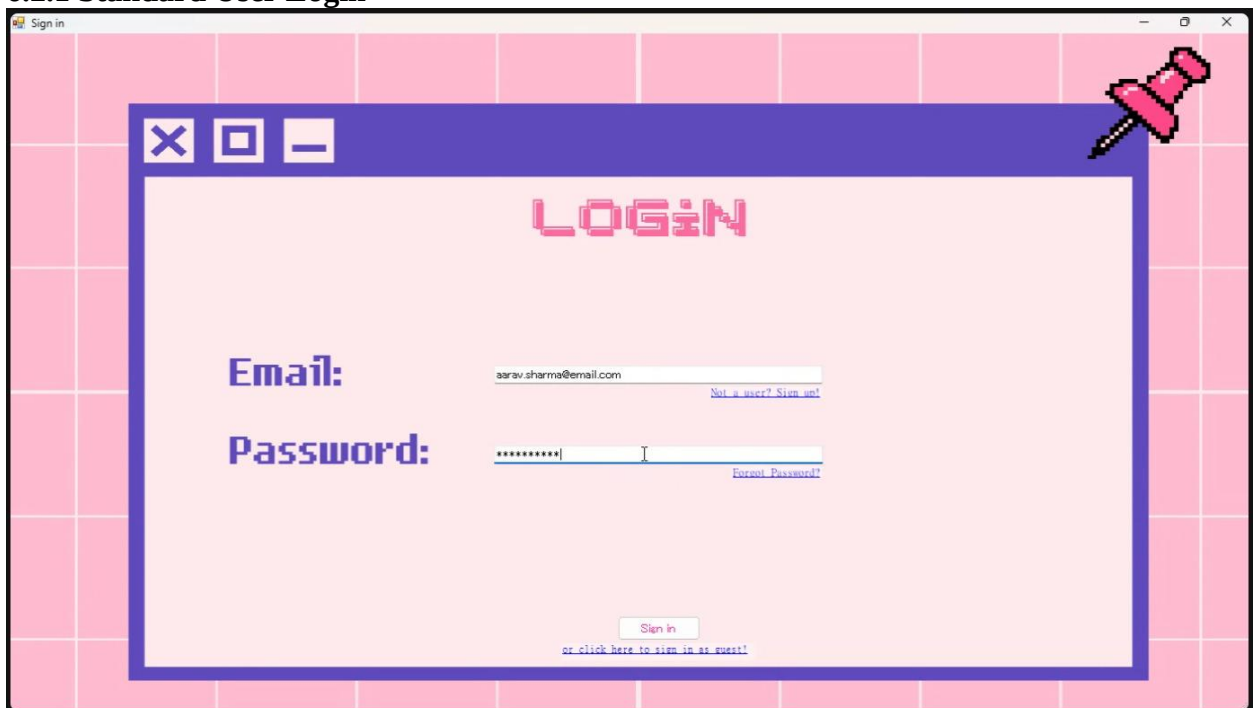


Figure 6.3: Standard User

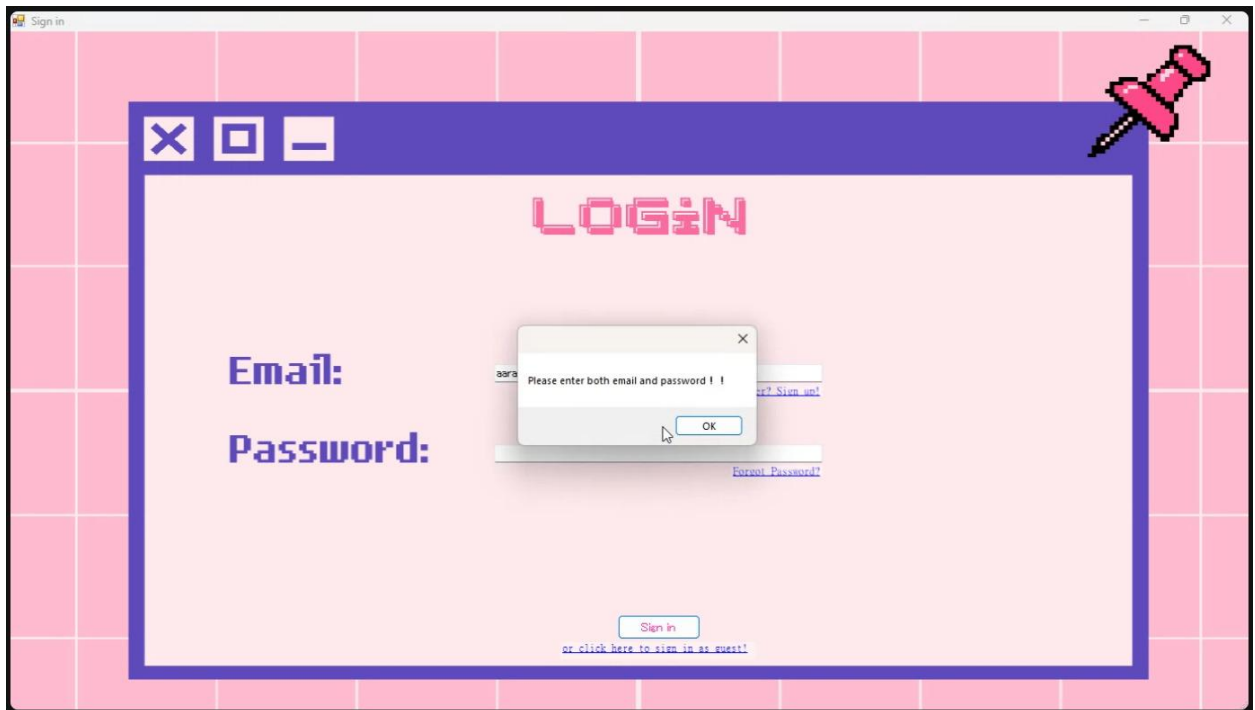


Figure 6.4: Error Handling for Login Page

6.2.2 Admin User Login

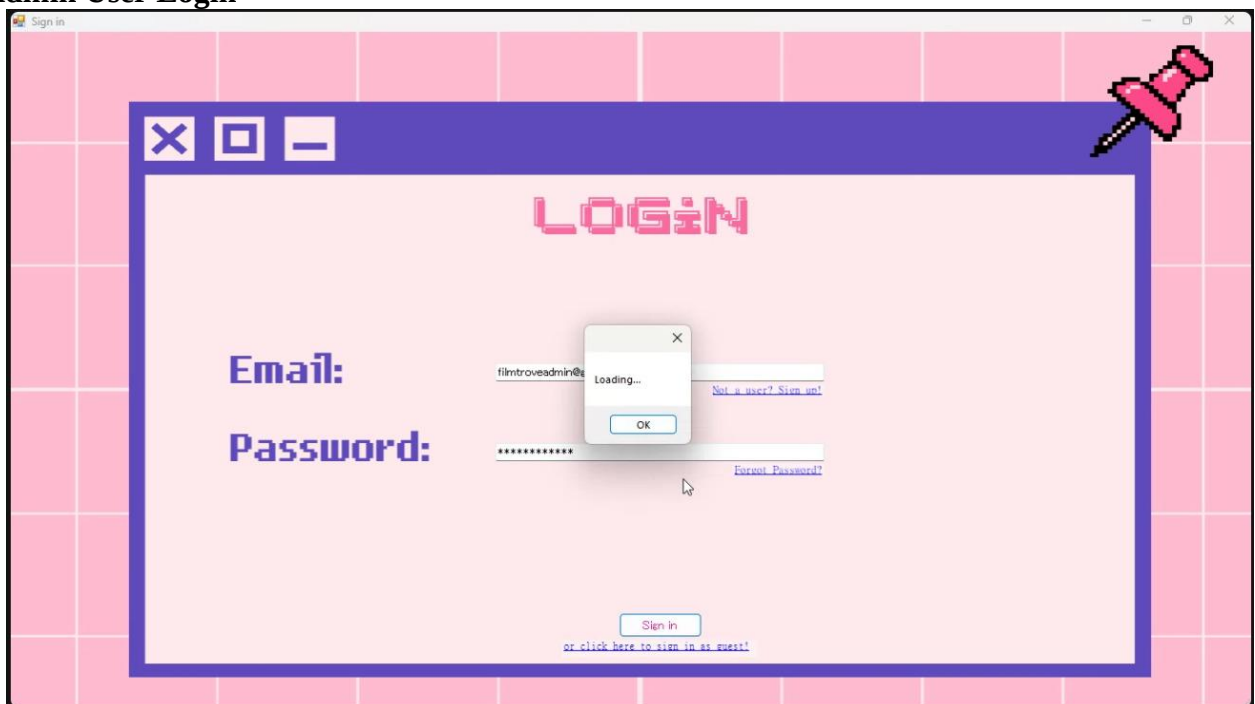
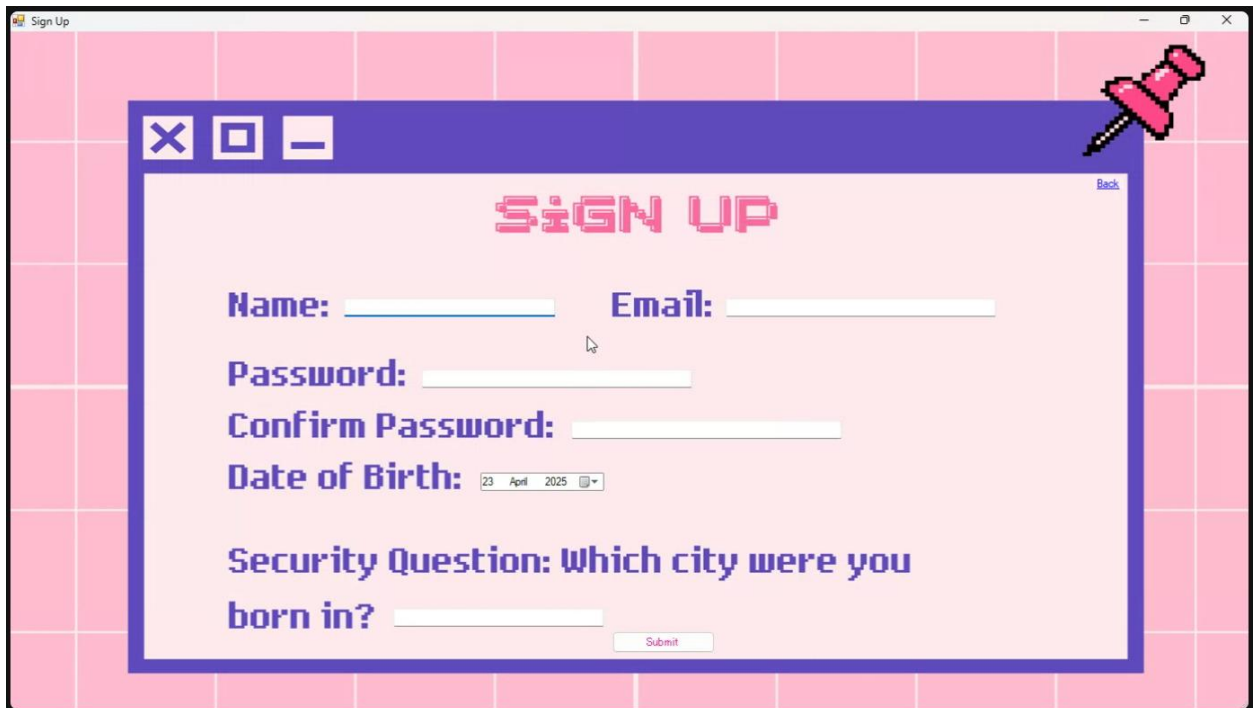


Figure 6.5: Admin Login

6.3 Sign Up Page



Sign Up

SIGN UP

[Back](#)

Name: Email:

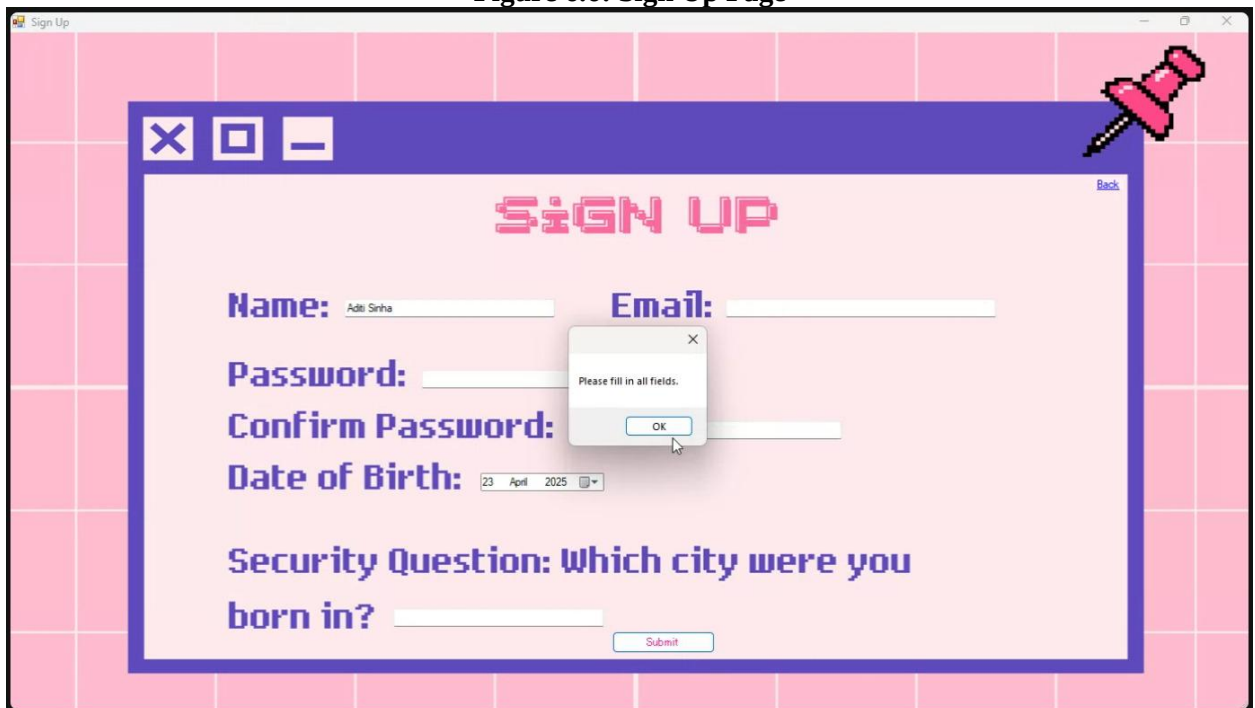
Password:

Confirm Password:

Date of Birth:

Security Question: Which city were you born in?

Figure 6.6: Sign Up Page



Sign Up

SIGN UP

[Back](#)

Name: Email:

Password:

Confirm Password:

Date of Birth:

Security Question: Which city were you born in?

Please fill in all fields.

Sign Up

SIGN UP

[Back](#)

Name: **Email:**

Password: **Confirm Password:**

Date of Birth:

Security Question: Which city were you born in?

Name must contain only letters.

OK

Sign Up

SIGN UP

[Back](#)

Name: **Email:**

Password: **Confirm Password:**

Date of Birth:

Security Question: Which city were you born in?

Please enter a valid email address.

OK

Sign Up

SIGN UP

[Back](#)

Name: **Email:**

Password: **Confirm Password:**

Date of Birth:

Security Question: Which city were you born in?

OK

Sign Up

SIGN UP

[Back](#)

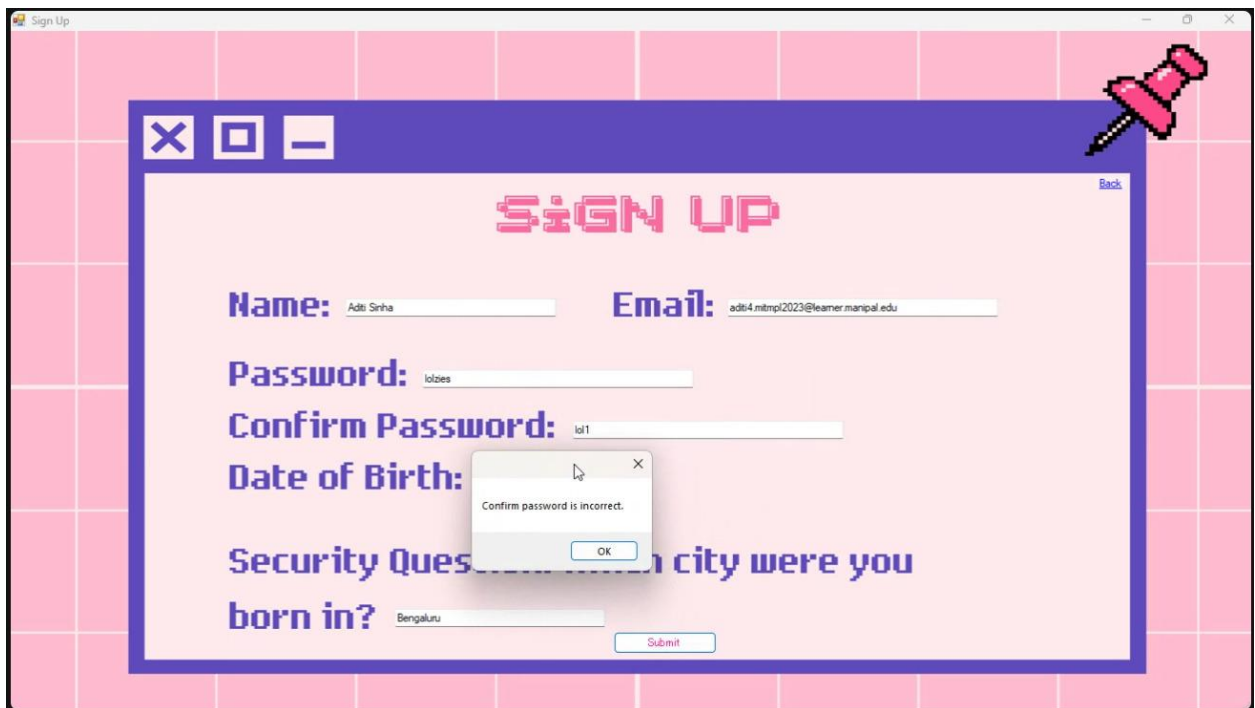
Name: **Email:**

Password: **Confirm Password:**

Date of Birth:

Security Question: Which city were you born in?

OK



A screenshot of a web browser window titled "Sign Up". The page has a pink grid background. A purple-bordered box contains the "SIGN UP" title in pink pixelated font. Below the title are five form fields: "Name:" (Aditi Sirha), "Email:" (aditi4.mitmpl2023@learner.manipal.edu), "Password:" (lolzies), "Confirm Password:" (lol1), and "Date of Birth:" (14 July 2004). A "Security Question: Which city were you born in?" field has "Bengaluru" entered. A "Submit" button is at the bottom right. A small error dialog box is open over the "Confirm Password" field, displaying the message "Confirm password is incorrect." with an "OK" button. A red pushpin is pinned to the top right corner of the browser window.

SIGN UP

Name: Aditi Sirha **Email:** aditi4.mitmpl2023@learner.manipal.edu

Password: lolzies **Confirm Password:** lol1

Date of Birth: 14 July 2004

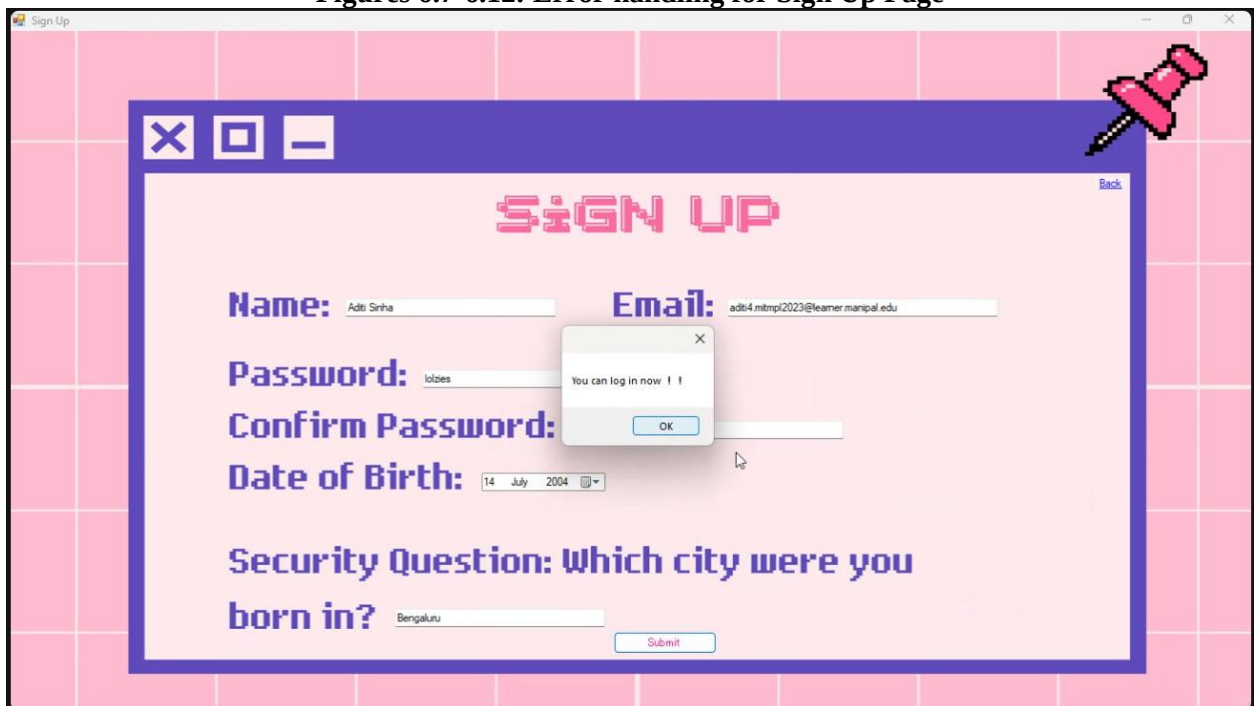
Security Question: Which city were you born in? Bengaluru

Submit

Confirm password is incorrect.

OK

Figures 6.7-6.12: Error handling for Sign Up Page



A screenshot of the same "Sign Up" web browser window. The form fields are the same as in the previous image. However, the "Confirm Password" field now contains "lolzies", matching the "Password" field. The error dialog box is replaced by a success dialog box that says "You can log in now !!". The "Submit" button is still present at the bottom right. The red pushpin remains in the top right corner.

SIGN UP

Name: Aditi Sirha **Email:** aditi4.mitmpl2023@learner.manipal.edu

Password: lolzies **Confirm Password:** lolzies

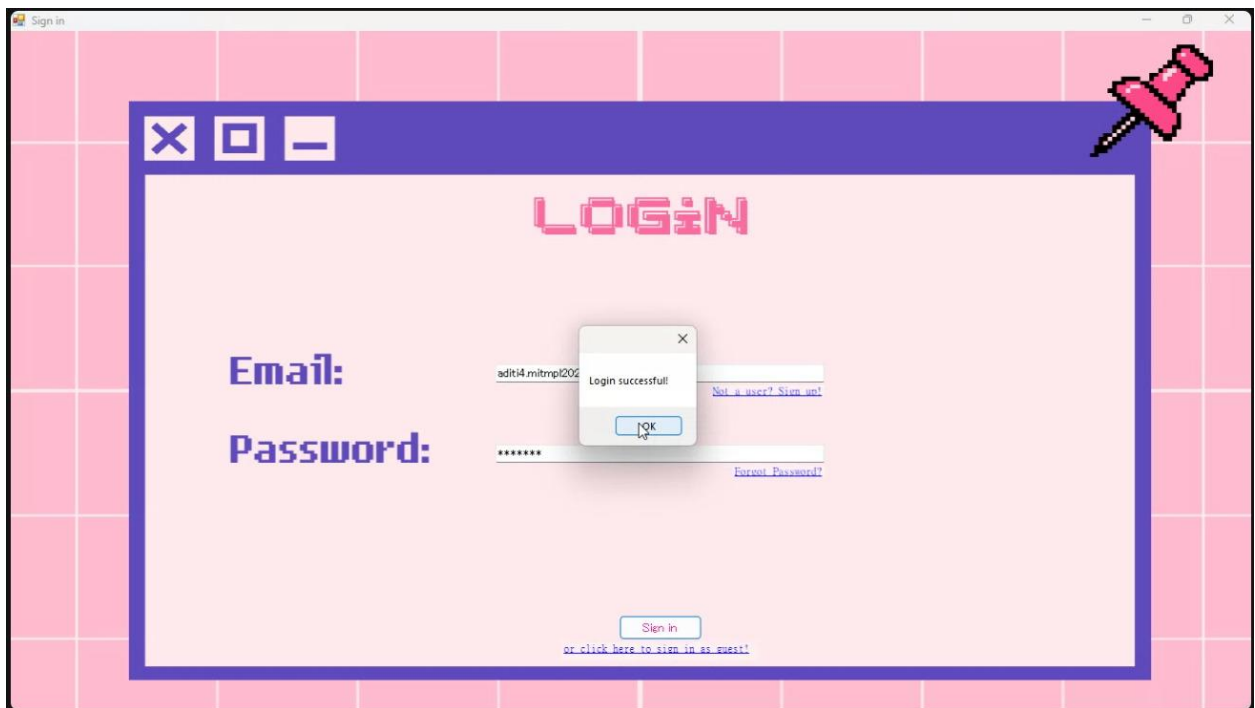
Date of Birth: 14 July 2004

Security Question: Which city were you born in? Bengaluru

Submit

You can log in now !!

OK



Figures 6.13-6.14: Successful Sign Up

6.4 User Dashboard

6.4.1 Standard User Dashboard

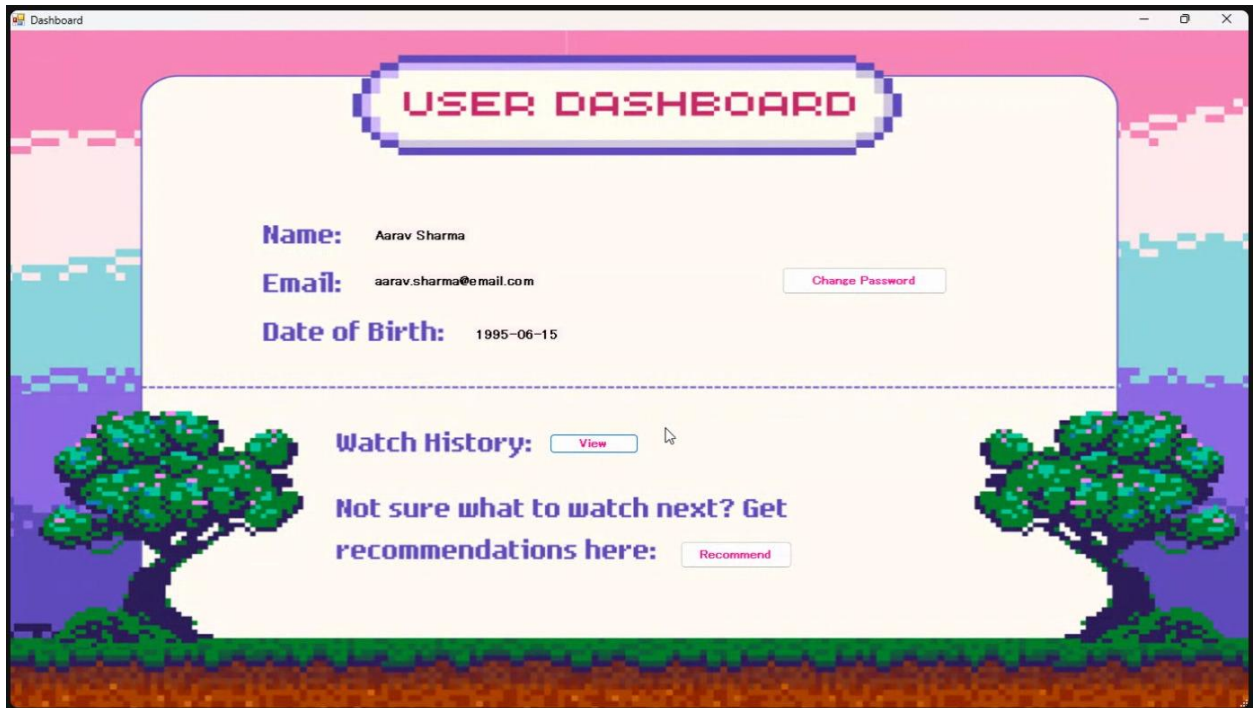


Figure 6.15: Standard User Dashboard

6.4.2 Guest User Dashboard

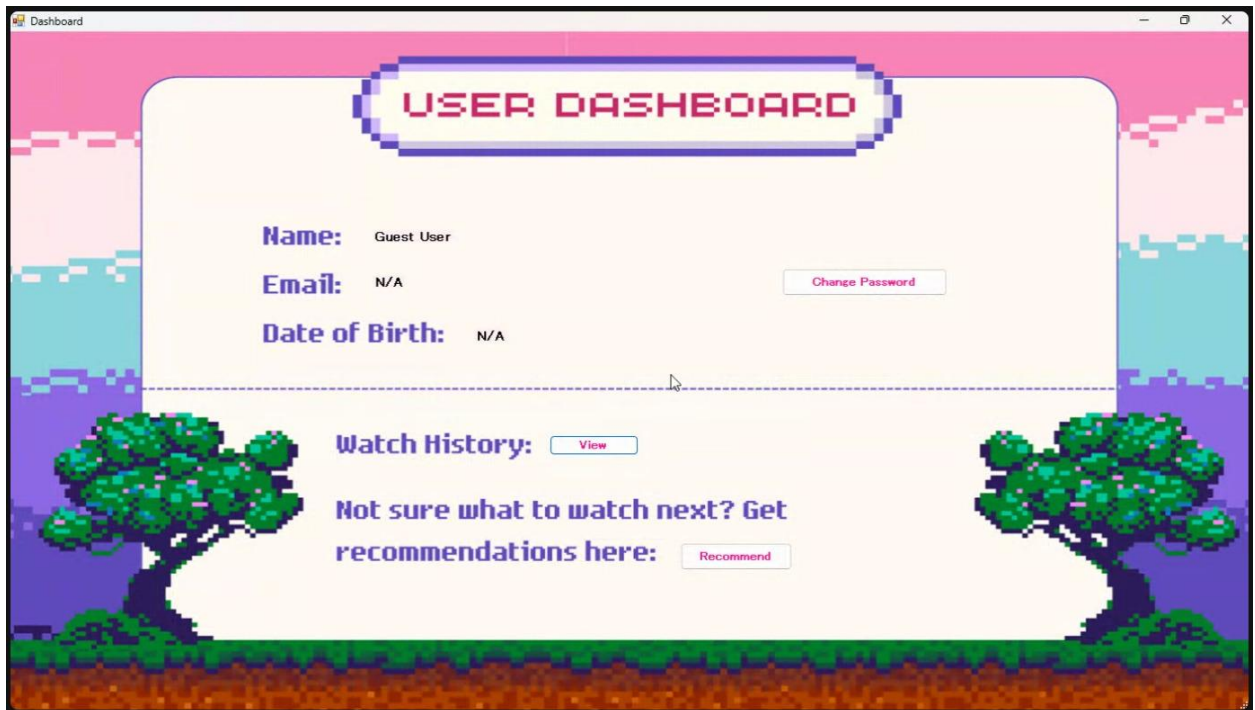


Figure 6.16: Guest User Dashboard

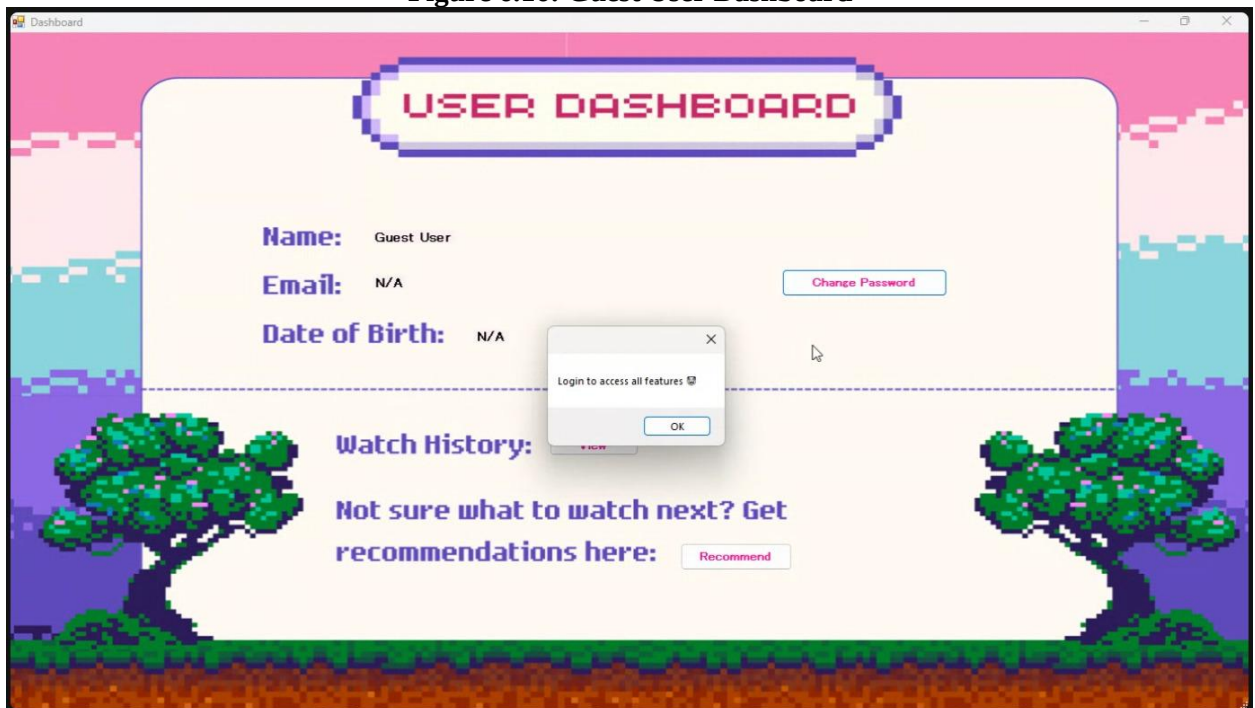


Figure 6.17: Guest users may only access Recommendation module

6.5 Watch History

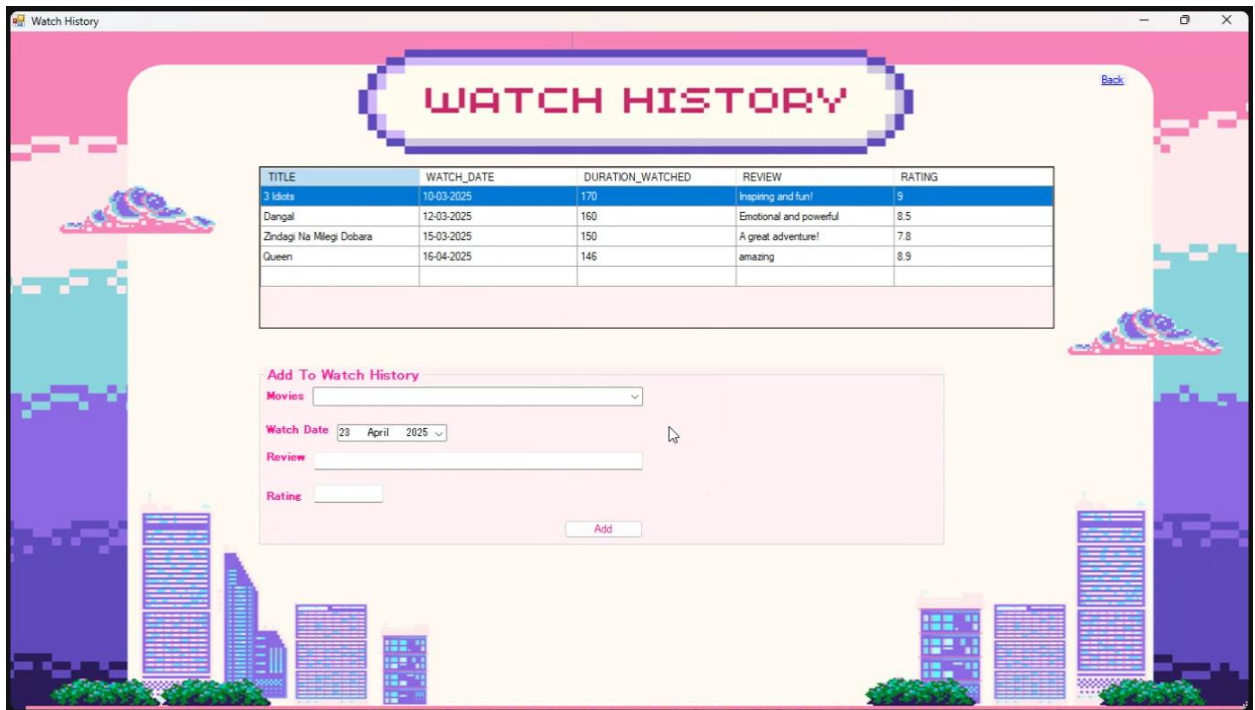


Figure 6.18: Watch History Page

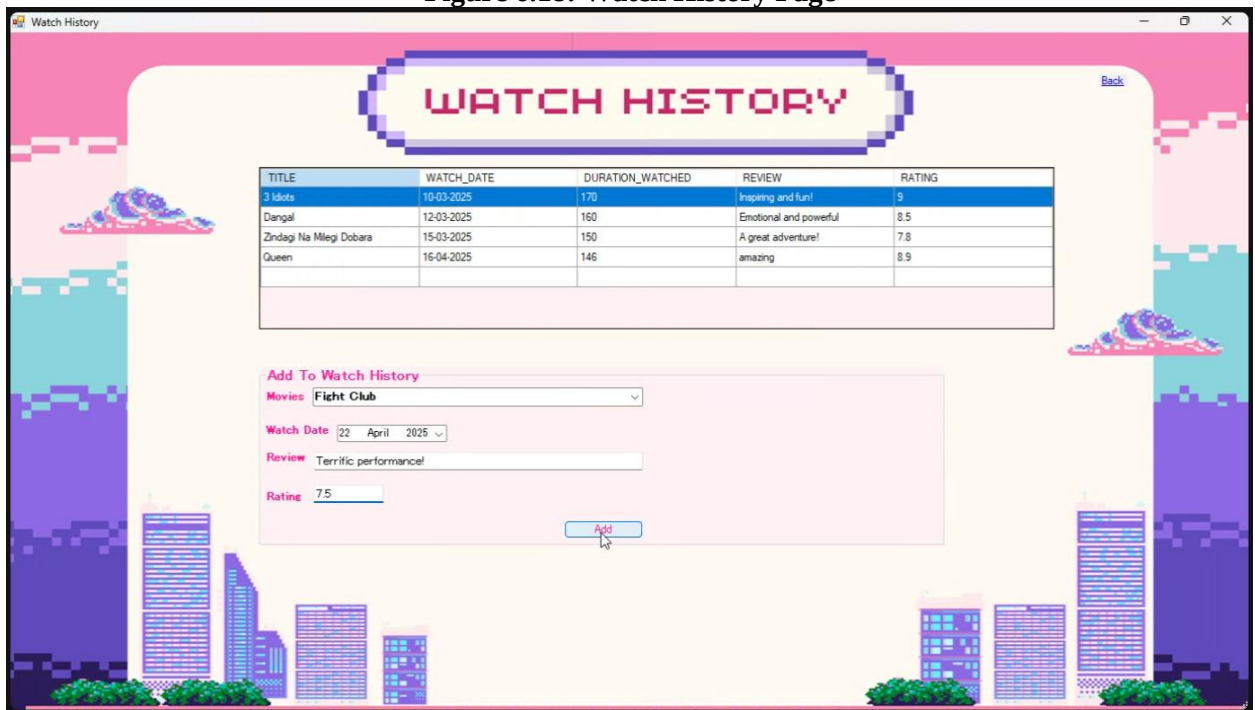


Figure 6.19: Review and Rating Module

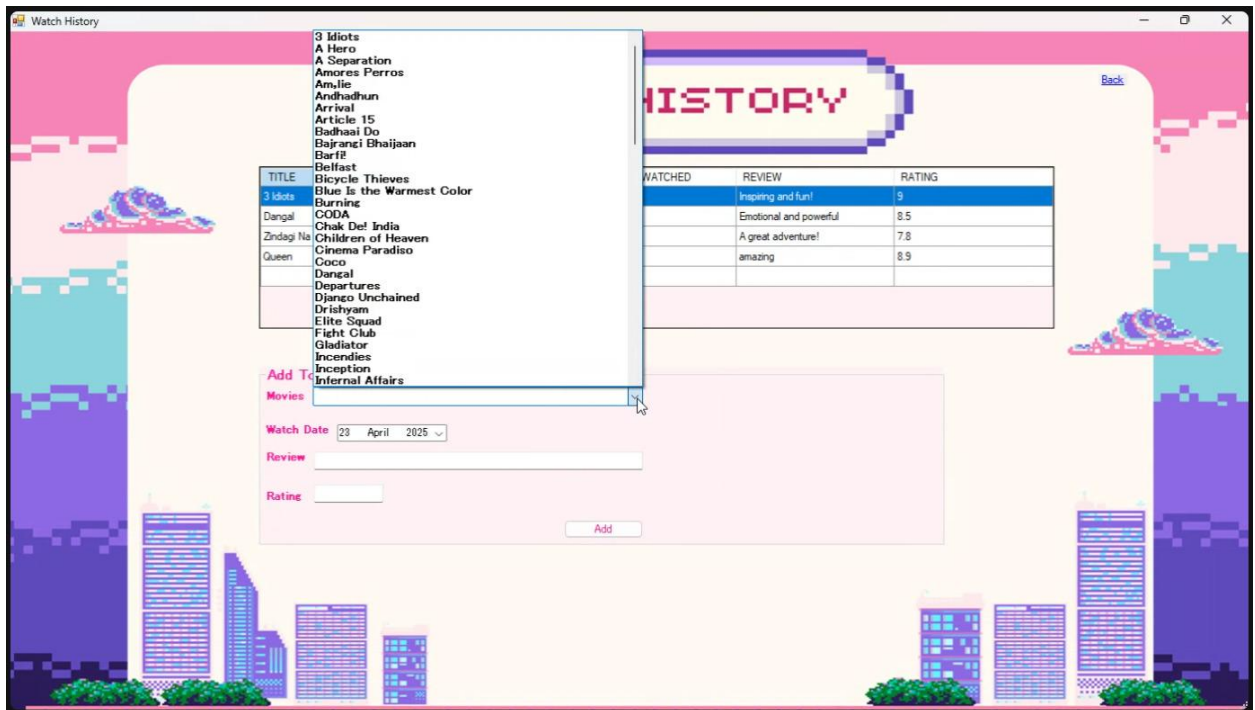


Figure 6.20: Movie Selection to add to Watch History

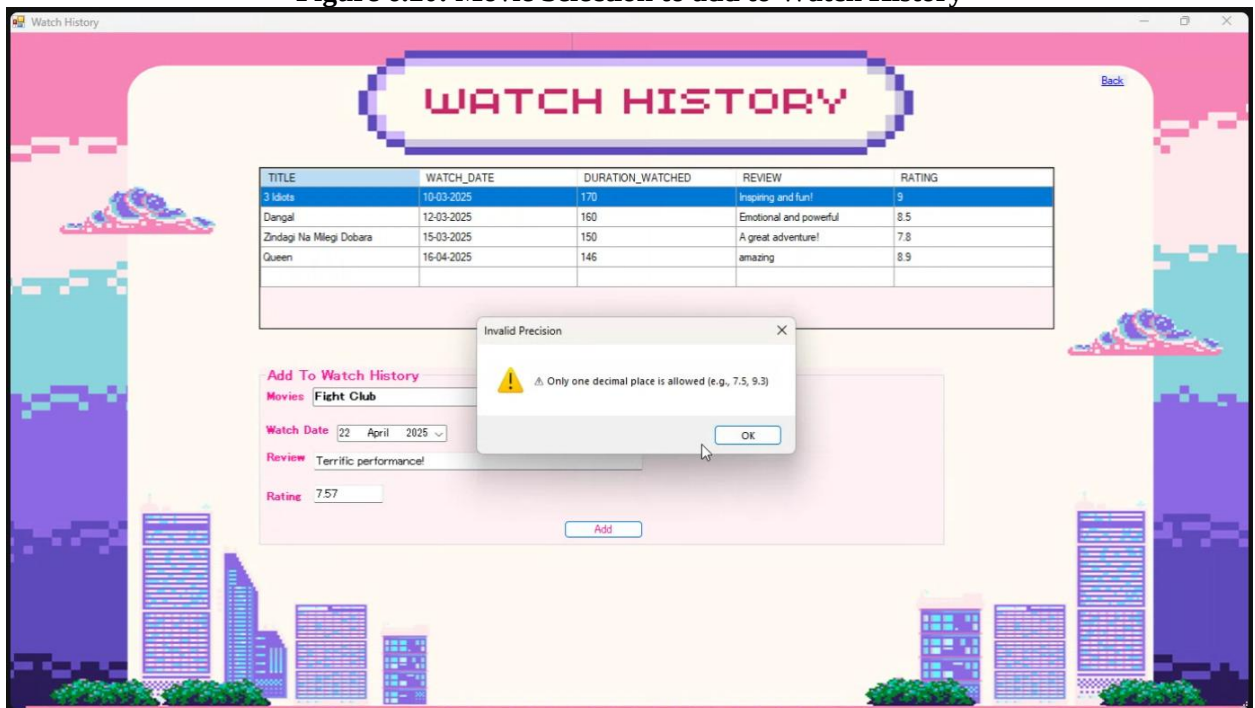


Figure 6.21: Error Handling on Watch History

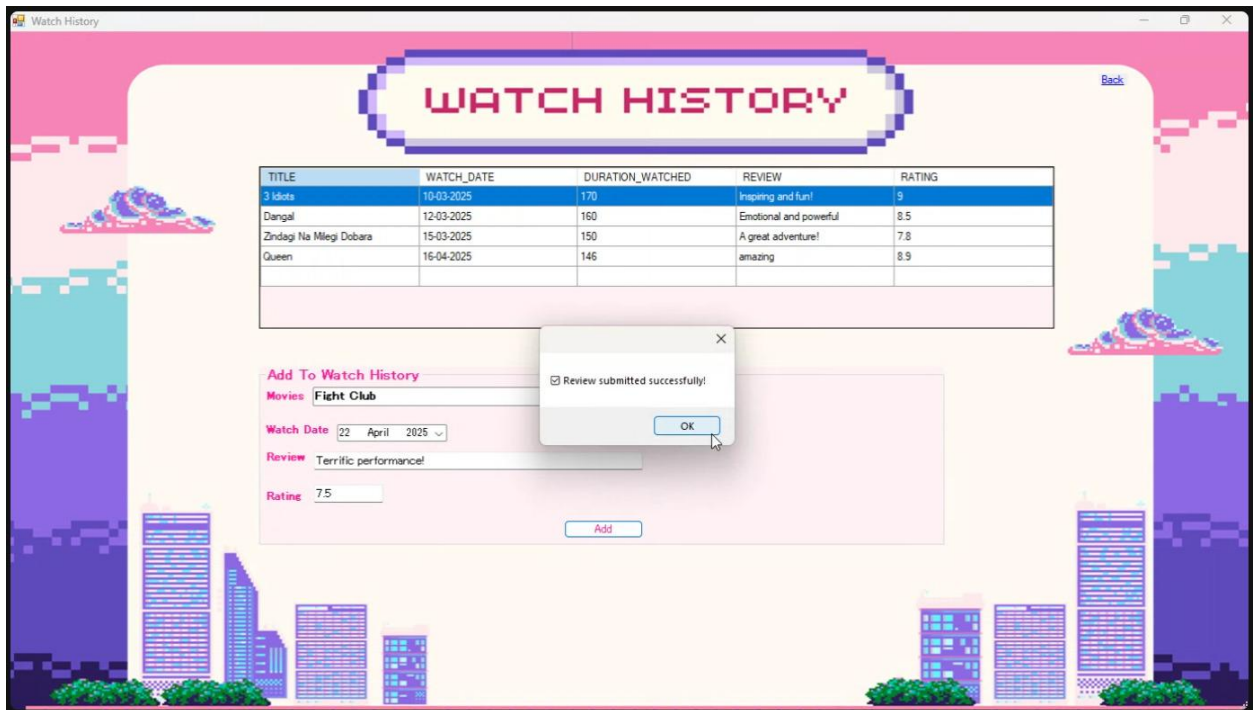


Figure 6.22: Review and Rating Submission

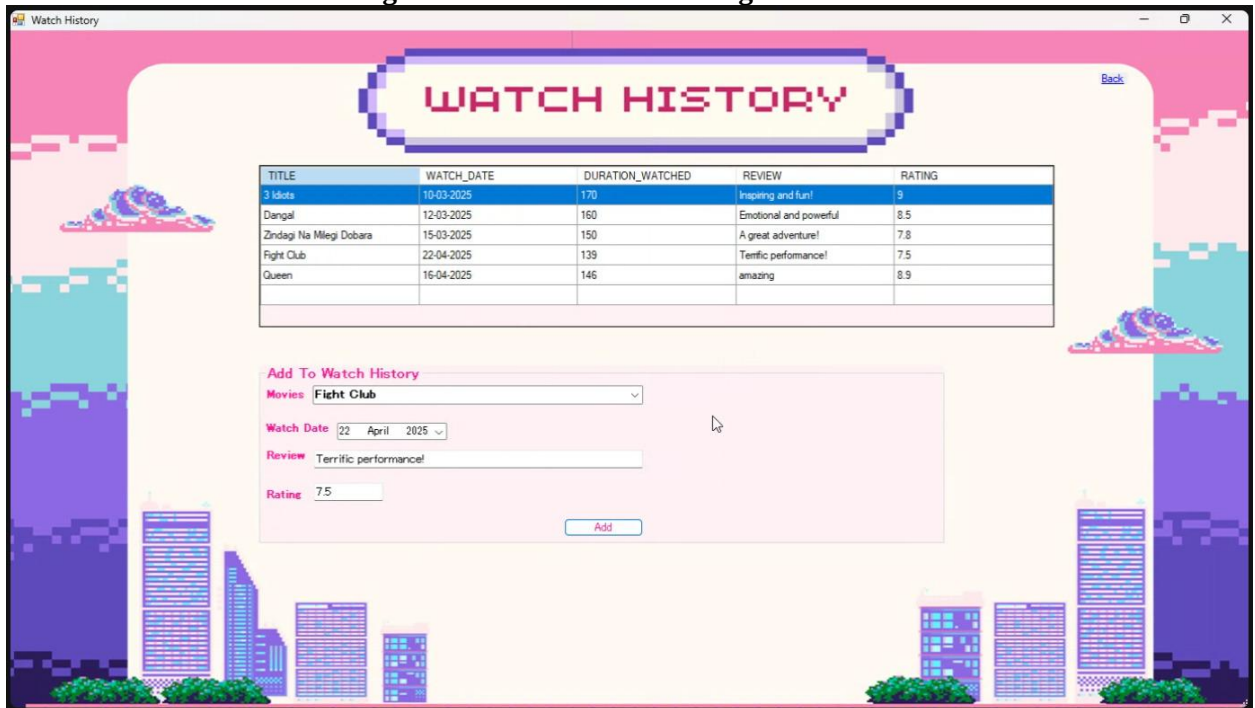


Figure 6.23: Updating Watch History Grid

6.6 Recommendation Page

6.6.1 Standard User Recommendation Page

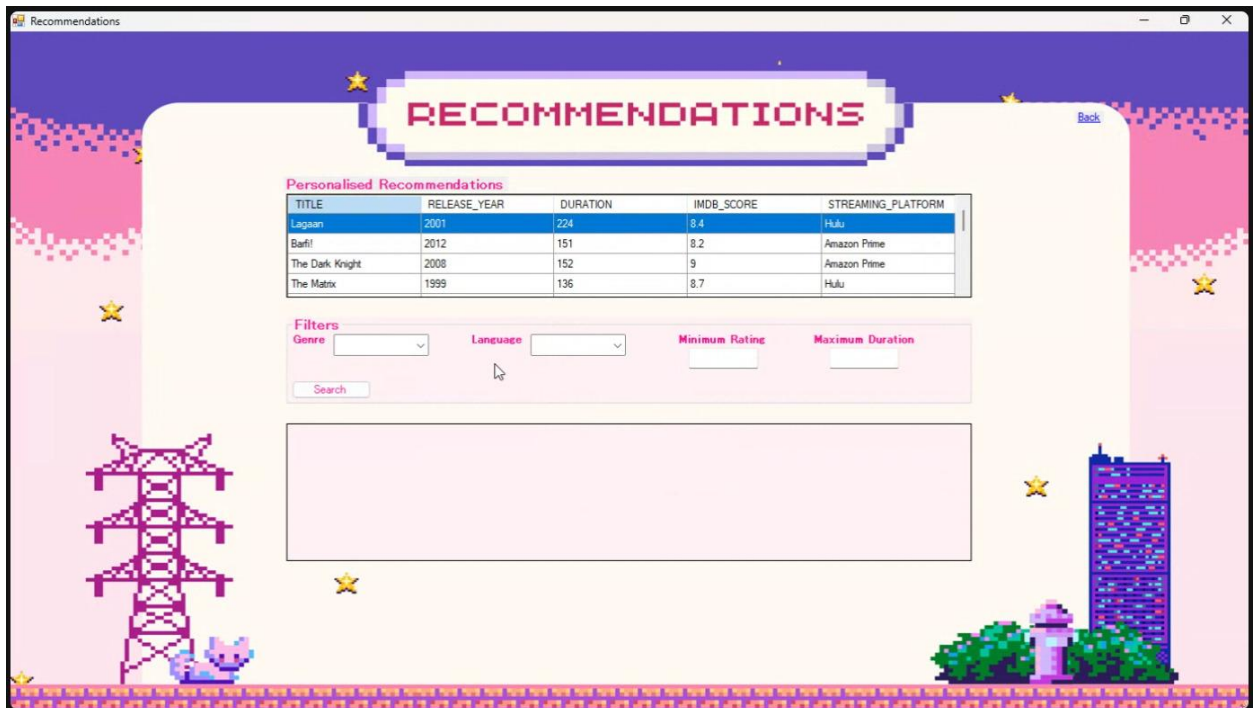


Figure 6.24: Personalised Recommendation based on Watch History

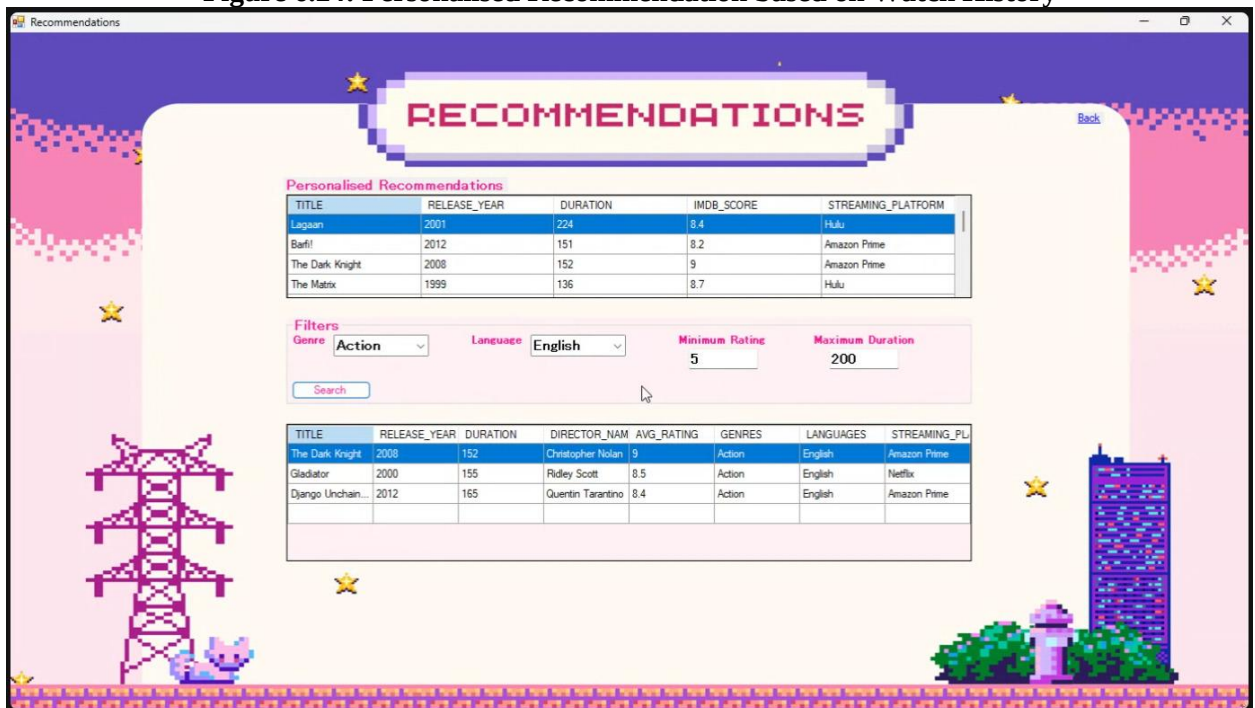


Figure 6.25: Search and Filter Module for Recommendations

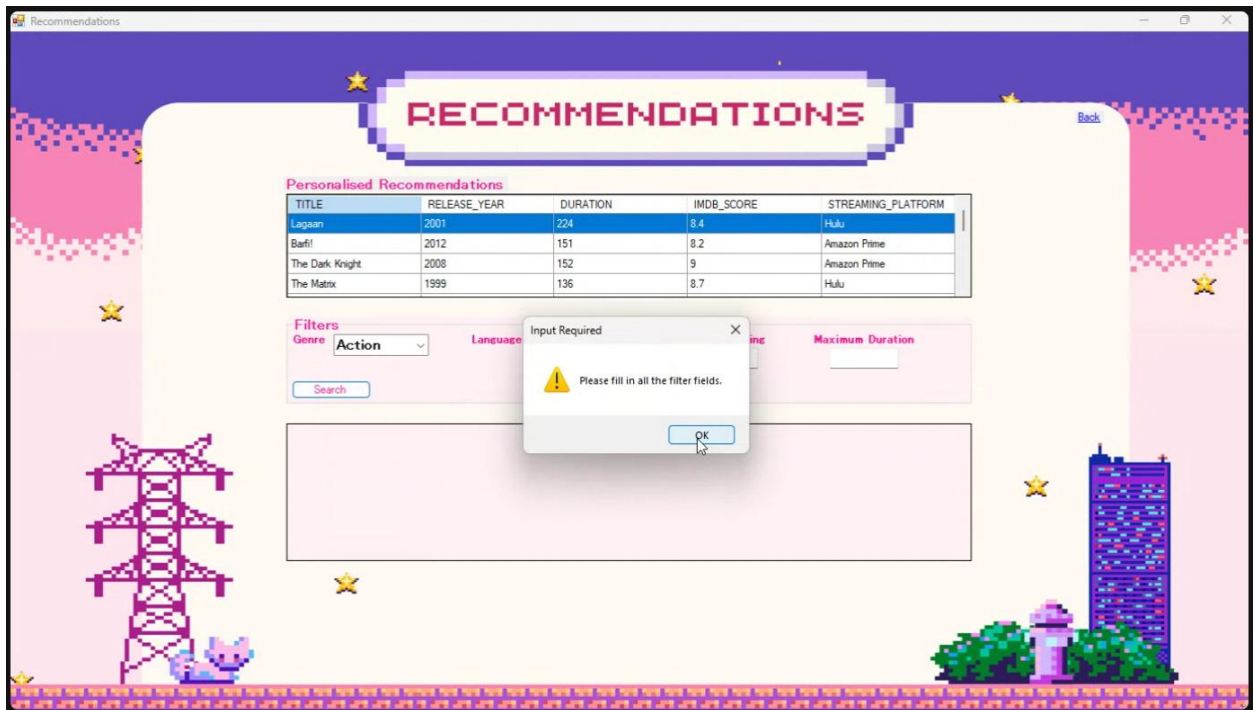


Figure 6.26: Error Handling for Recommendation Page

6.6.2 Guest User Recommendation Page

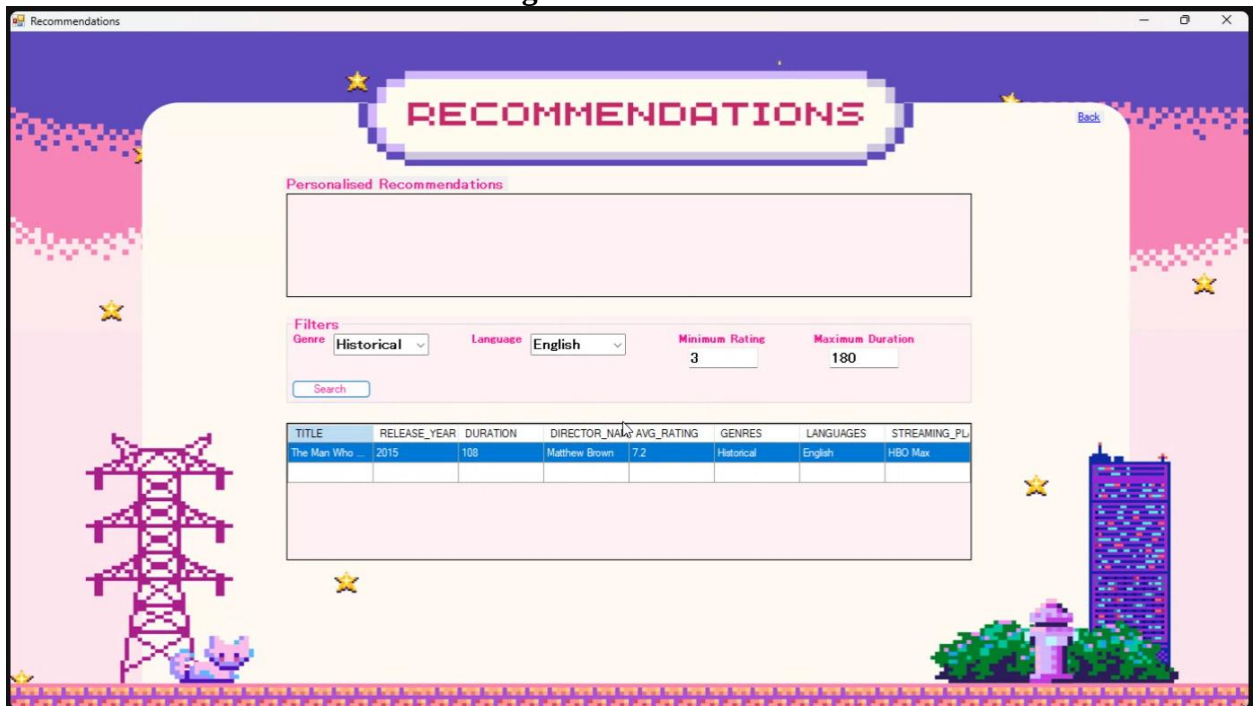
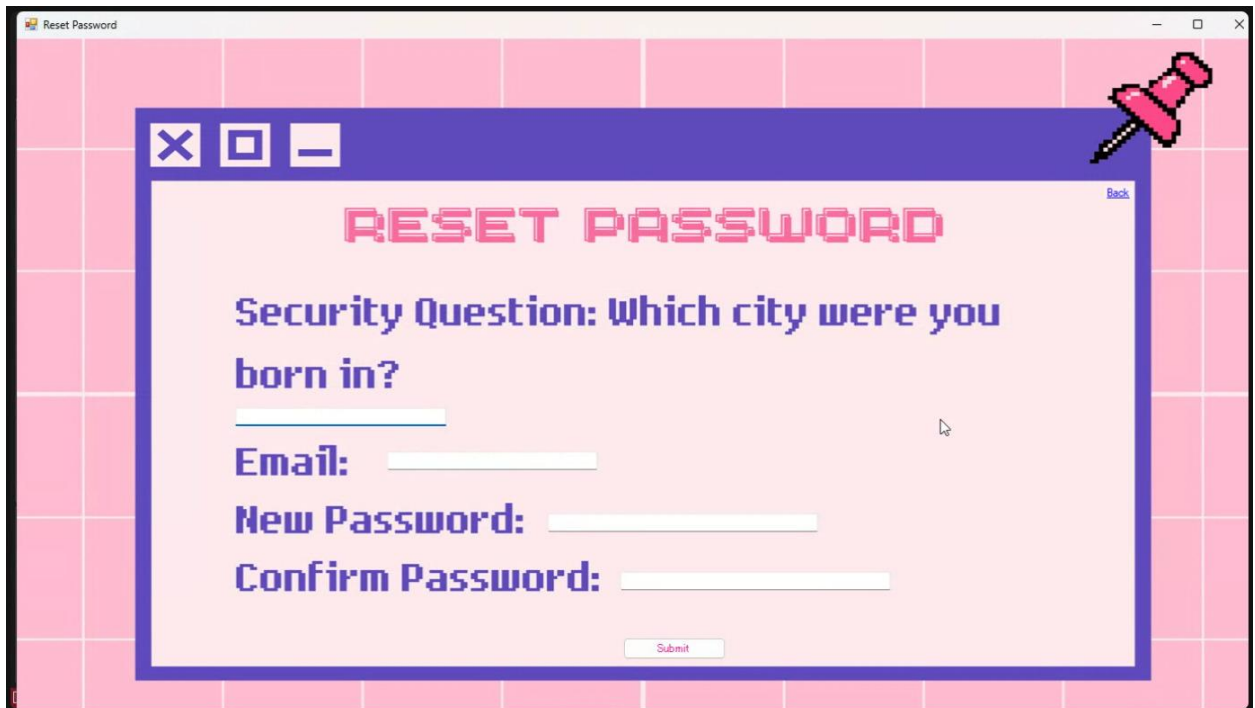


Figure 6.27: Limited Recommendation Module for Guest User without Personalised Recommendations

6.7 Forgot/Reset Password Page



The screenshot shows a web browser window titled "Reset Password". The page has a pink grid background. A blue-bordered box contains the title "RESET PASSWORD" in pink, a security question "Security Question: Which city were you born in?" in blue, and three input fields for "Email:", "New Password:", and "Confirm Password:". A "Submit" button is at the bottom. A "Back" link is in the top right of the blue box. A red pushpin is in the top right corner of the browser window.

Reset Password

RESET PASSWORD

Security Question: Which city were you born in?

Email:

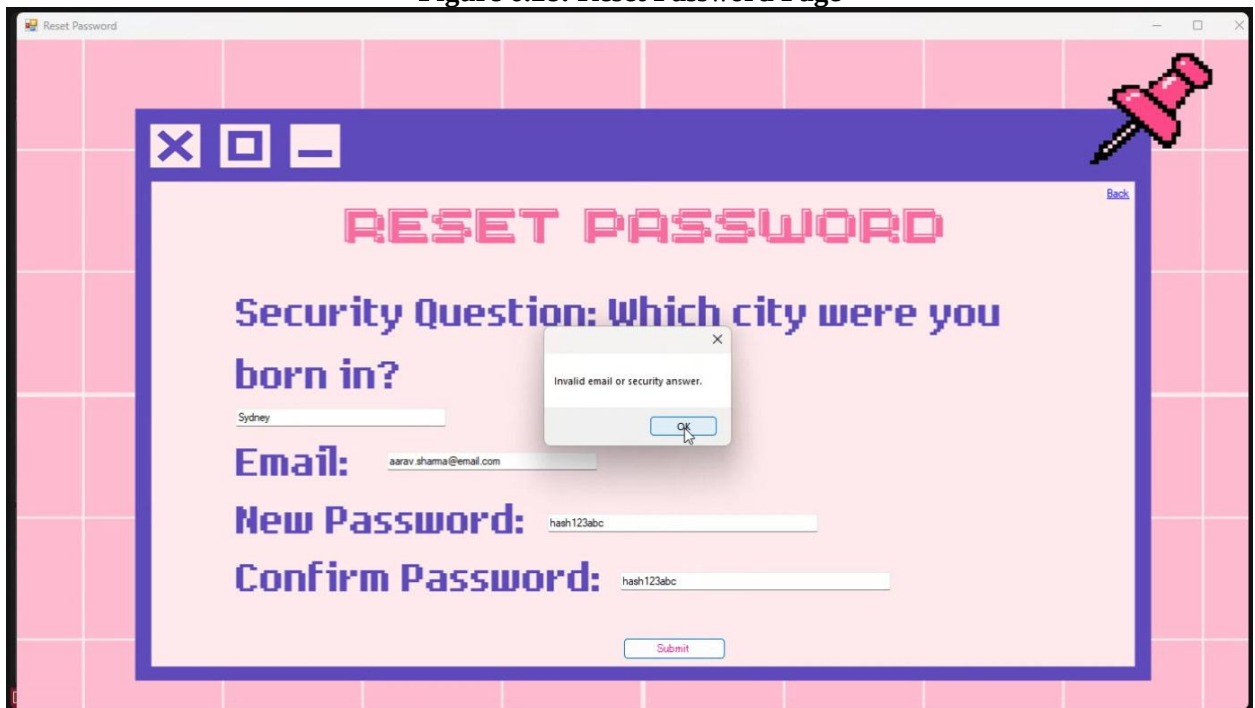
New Password:

Confirm Password:

Submit

Back

Figure 6.28: Reset Password Page



The screenshot shows the same "Reset Password" page as Figure 6.28, but with an error dialog box displayed. The dialog box has a title bar with a close button (X) and contains the text "Invalid email or security answer." and an "OK" button. The input fields now contain the values "Sydney", "aarav.sharma@email.com", "hash123abc", and "hash123abc".

Reset Password

RESET PASSWORD

Security Question: Which city were you born in?

Sydney

Email: aarav.sharma@email.com

New Password: hash123abc

Confirm Password: hash123abc

Submit

Invalid email or security answer.

OK

Figure 6.29: Error Handling for Reset Password Page

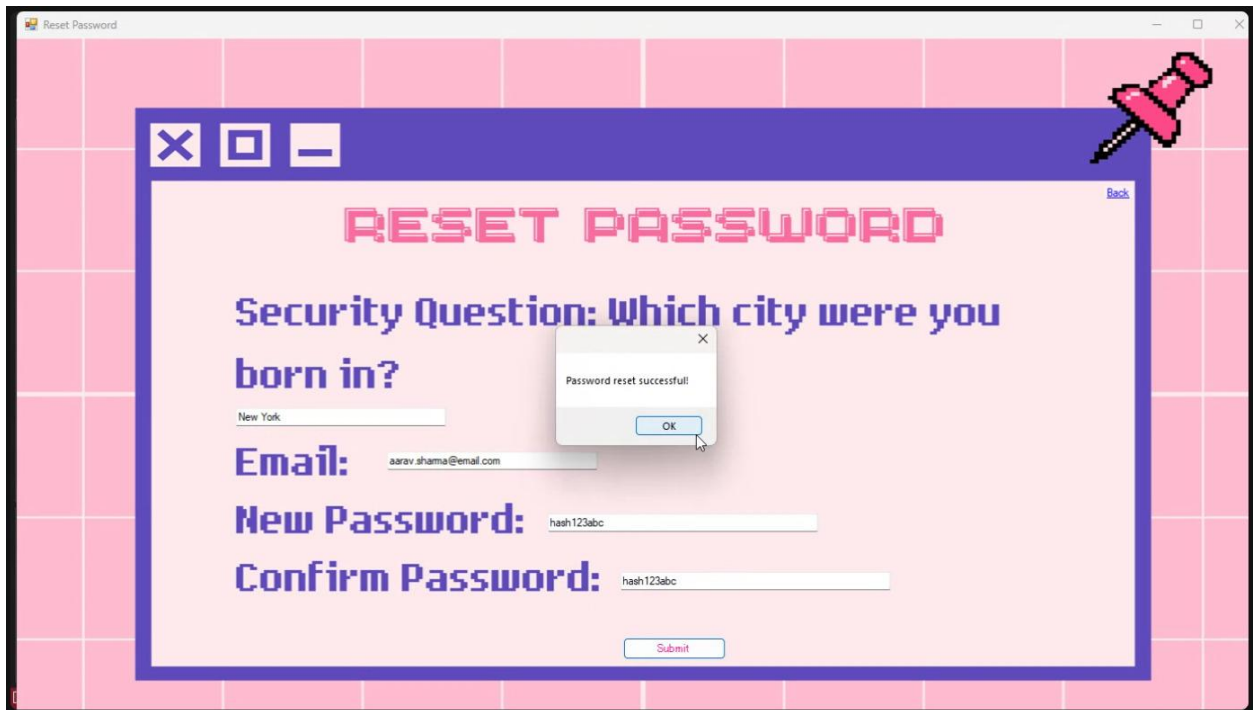


Figure 6.30: Successfully Reset Password

6.8 Admin Access Page

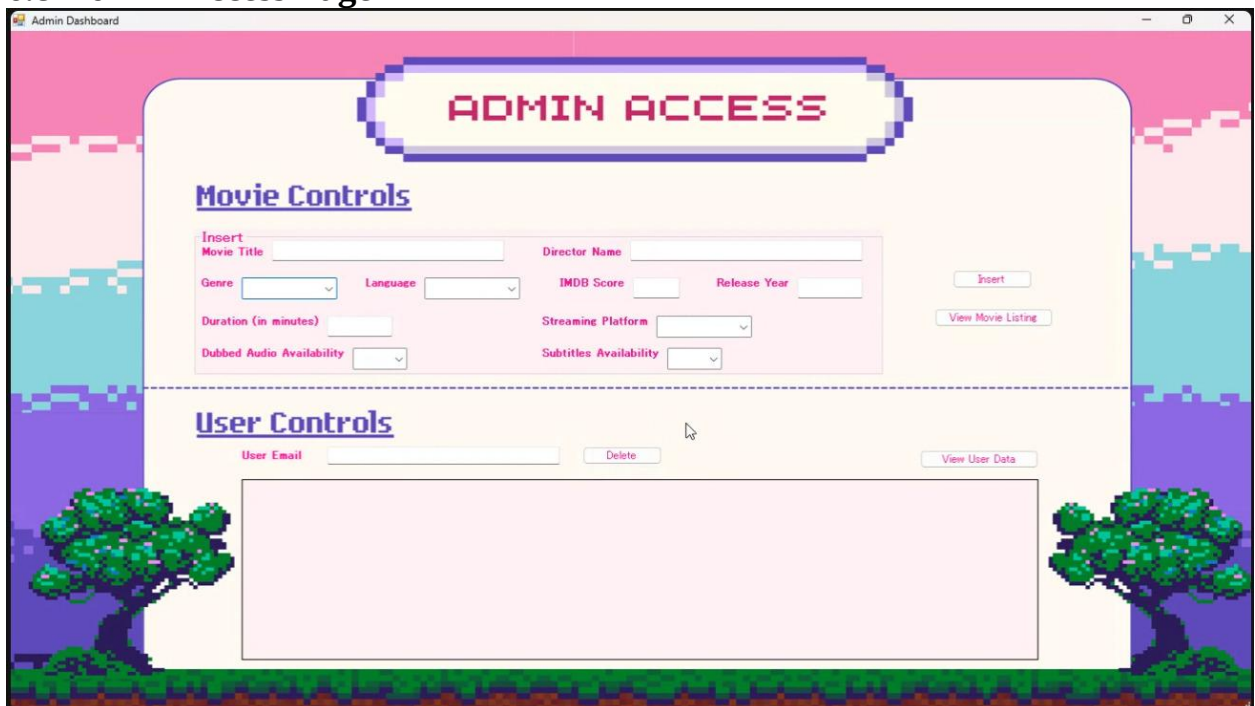


Figure 6.31: Admin Dashboard

6.8.1 Insert Movie Permission



Figure 6.32-6.33: Insert movie into database

6.8.2 View Movie Database

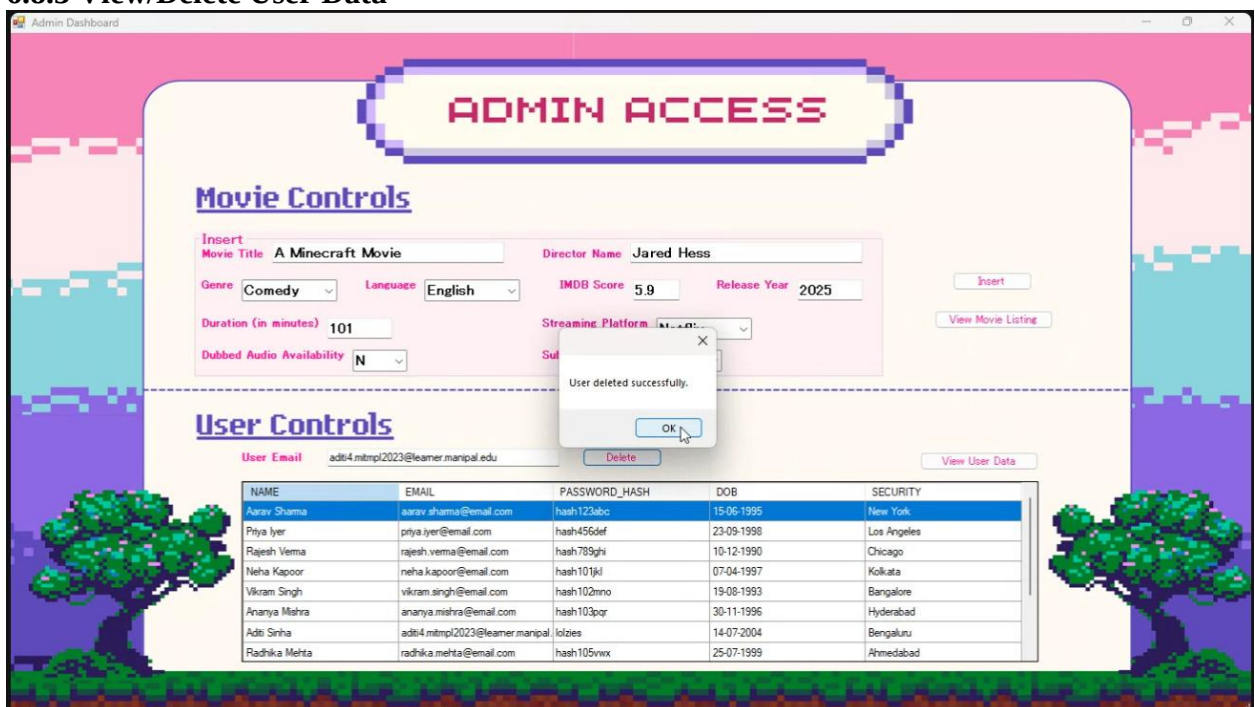


The screenshot shows a web application window titled "Movie List". It features a pixelated background with a pink sky, blue water, and a purple ground. A large, stylized "MOVIE LISTINGS" title is at the top. Below it is a table with 8 columns: TITLE, RELEASE_YEAR, DURATION, IMDB_SCORE, DIRECTOR_NAME, GENRE_NAME, LANG_NAME, and STR_NAME. The table contains 20 rows of movie data. A "Back" button is visible in the top right corner of the table area.

TITLE	RELEASE_YEAR	DURATION	IMDB_SCORE	DIRECTOR_NAME	GENRE_NAME	LANG_NAME	STR_NAME
3 Idiots	2009	170	8.4	Rajkumar Hirani	Comedy	Hindi	Netflix
A Hero	2021	127	7.5	Aghar Farhadi	Crime	Persian	Apple TV+
A Separation	2011	123	8.4	Aghar Farhadi	Drama	Persian	Hulu
Amélie	2001	122	8.3	Jean-Pierre Jeunet	Romance	French	Netflix
Amores Perros	2000	154	8.1	Alejandro Gonz lez	Crime	Spanish	Disney+
Andhadhun	2018	139	8.5	Sriram Raghavan	Thriller	Hindi	Hulu
Arrival	2016	116	7.9	Denis Villeneuve	Sci-Fi	English	Amazon Prime
Article 15	2019	130	8.1	Anubhav Sinha	Crime	Hindi	Disney+
Badhaai Do	2022	147	7.3	Harshvardhan	Comedy	Hindi	HBO Max
Bajrangi Bhaijaan	2015	163	8.1	Kabir Khan	Drama	Hindi	HBO Max
Barfi!	2012	151	8.2	Anurag Basu	Comedy	Hindi	Amazon Prime
Belfast	2021	98	7.3	Kenneth Branagh	Drama	English	Disney+
Bicycle Thieves	1948	89	8.3	Vittorio De Sica	Historical	Italian	Netflix
Blue Is the Warmest Color	2013	180	7.7	Abdellatif Kechiche	Romance	French	Amazon Prime
Burning	2018	148	7.5	Lee Chang-dong	Thriller	Korean	Amazon Prime
Chak De! India	2007	153	8.2	Shimit Amin	Sports	Hindi	HBO Max
Children of Heaven	1997	89	8.3	Majid Majidi	Drama	Persian	Apple TV+
Cinema Paradiso	1988	155	8.5	Giuseppe Tornatore	Historical	Italian	Apple TV+
Concussion	2015	135	7.3	John Singleton	Drama	English	HBO Max

Figure 6.34: Movie Listings View

6.8.3 View/Delete User Data



The screenshot shows an "Admin Dashboard" window. It has a pixelated background with a pink sky, blue water, and purple ground. The main title is "ADMIN ACCESS". Below it, there are two sections: "Movie Controls" and "User Controls".

Movie Controls

Insert
 Movie Title: Director Name:
 Genre: Language: IMDB Score: Release Year:
 Duration (in minutes): Streaming Platform:
 Dubbed Audio Availability: Subtitle Availability:
 Buttons:

User Controls

User Email:

User Data Table

NAME	EMAIL	PASSWORD_HASH	DOB	SECURITY
Aarav Sharma	aarav.sharma@gmail.com	hash123abc	15-06-1995	New York
Priya Iyer	priya.iyer@gmail.com	hash456def	23-09-1998	Los Angeles
Rajesh Verma	rajesh.verma@gmail.com	hash789ghi	10-12-1990	Chicago
Neha Kapoor	neha.kapoor@gmail.com	hash101kl	07-04-1997	Kolkata
Vikram Singh	vikram.singh@gmail.com	hash102mno	19-08-1993	Bangalore
Ananya Mishra	ananya.mishra@gmail.com	hash103pqr	30-11-1996	Hyderabad
Aditi Sinha	aditi4.mtmpl2023@leamer.manipal.edu	lolzies	14-07-2004	Bengaluru
Radhika Mehta	radhika.mehta@gmail.com	hash105vwx	25-07-1999	Ahmedabad

Figure 6.35: Delete specific user data by email

Admin Dashboard

ADMIN ACCESS

Movie Controls

Insert

Movie Title

A Minecraft Movie

Director Name

Jared Hess

Genre

Comedy

Language

English

IMDB Score

5.9

Release Year

2025

Duration (in minutes)

101

Streaming Platform

Netflix

Dubbed Audio Availability

N

Subtitles Availability

Y

Insert

View Movie Listing

User Controls

User Email

adi4.mttrpl2023@learner.manipal.edu

Delete

View User Data

NAME	EMAIL	PASSWORD_HASH	DOB	SECURITY
Aarav Sharma	aarav.sharma@email.com	hash123abc	15-06-1995	New York
Priya Iyer	priya.iyer@email.com	hash456def	23-09-1998	Los Angeles
Rajesh Vema	rajesh.vema@email.com	hash789ghi	10-12-1990	Chicago
Neha Kapoor	neha.kapoor@email.com	hash101kl	07-04-1997	Kolkata
Vikram Singh	vikram.singh@email.com	hash102mno	19-08-1993	Bangalore
Ananya Mishra	ananya.mishra@email.com	hash103pqr	30-11-1996	Hyderabad
Radhika Mehta	radhika.mehta@email.com	hash105vwx	25-07-1999	Ahmedabad
Sandeep Reddy	sandeep.reddy@email.com	hash106yz	05-10-1991	Jaipur

Figure 6.36: View User Database

Chapter 7

Conclusion and Future Work

1.1 Conclusion

FilmTrove, as an intelligent user-driven movie recommendation system, was meticulously crafted and developed to encompass a modern database management system capturing the essence of agile content discovery. It integrates several features like user registration, movie search, watch history, personalized recommendations, and even administrative moderation into a single friendly interface for movies.

FilmTrove maintains data integrity, consistency, and optimal query execution with its modular backend .NET Core structure and modern web frontend's cross-device, real-time interaction with the platform. FilmTrove uses logic-based recommendations via SQLPlus and normalized relational databases to achieve this.

The responsive UI is systematically arranged and provides secured pathways to content from both user and admin perspectives. This allows straightforward management and exploration for film enthusiasts. FilmTrove adeptly curates a customized experience by analyzing metadata, ratings, watch history, and user profiles for enhanced dynamic discovery, all without a discrete recommendation engine.

In the field of entertainment, this undertaking illustrates the possibilities of systematized database-driven solutions tailoring intelligent, user-specific content.

1.2 Future Work

While FilmTrove fulfils core movie recommendation and user management functionalities, there are several avenues for enhancement that can elevate the platform's intelligence, scalability, and user engagement. Future iterations could consider the following improvements:

- **Refined Recommendations:** Enhancing the search and filter system by integrating multiple entities.
- **AI Recommendations:** Use machine learning to improve personalization.
- **Email Notifications:** Alerts for recommendations, reviews, and account updates.
- **Streaming Integration:** Direct links to Netflix, Prime Video, etc.
- **Smart TV Support:** Cast and stream directly to smart devices.
- **Cloud Hosting:** Improve scalability and uptime.
- **Language Support:** Add multilingual and localized content.
- **Admin Analytics:** Dashboards to track usage and trends.
- **Premium Features:** Ad-free experience, curated picks, or early access.

By incorporating these enhancements, FilmTrove could evolve into a full-featured, scalable, and intelligent media platform, offering users a richer and more personalized entertainment experience.

Chapter 8

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7. Scribd, Document Sharing and Publishing Platform, Available: <https://www.scribd.com>

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